Chapter XI. Modular Form

1. Basic Definition

1.1. Action on the extended complex plane. $\mathrm{GL}_2(\mathbb{C})$ acts on $\widehat{\mathbb{C}}$. For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $z \in \widehat{\mathbb{C}}$,

$$gz \coloneqq g \cdot z = \begin{cases} \dfrac{az+b}{cz+d}, & z \in \mathbb{C},\ cz+d \neq 0, \\ \infty, & z \in \mathbb{C},\ cz+d = 0, \\ \infty, & z = \infty,\ c = 0, \\ \dfrac{a}{c}, & z = \infty,\ c \neq 0. \end{cases}$$

$$Z(\mathrm{GL}_2(\mathbb{C})) = \left\{ \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} : a \in \mathbb{C}^\times \right\}.$$

$$\mathcal{H} = \{z \in \mathbb{C} : \mathrm{Im}(z) > 0\}, \qquad \text{upper half-plane.}$$

Proposition 11.1.1 *Let $g \in \mathrm{GL}_2(\mathbb{R})$ and $z \in \mathbb{C}$. If $cz + d \neq 0$, then*

$$\mathrm{Im}(gz) = \frac{\det(g)}{|cz+d|^2}\mathrm{Im}(z).$$

Proof.

$$\begin{aligned} \mathrm{Im}(gz) &= \mathrm{Im}\left(\frac{(az+b)(c\bar{z}+d)}{(cz+d)(c\bar{z}+d)}\right) \\ &= \mathrm{Im}\left(\frac{acz\bar{z} + bd + adz + bc\bar{z}}{|cz+d|^2}\right) \\ &= \frac{ad\mathrm{Im}(z) - bc\mathrm{Im}(z)}{|cz+d|^2} \\ &= \frac{(ad-bc)\mathrm{Im}(z)}{|cz+d|^2}. \end{aligned}$$

□

Lemma 11.1.2 *If $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{R})$ and $z \in \mathbb{C} \setminus \mathbb{R}$, then $cz + d \neq 0$.*

because $\det(g) \neq 0 \Longrightarrow (c,d) \neq (0,0)$.

Corollary 11.1.3 *The action of $\mathrm{GL}_2^+(\mathbb{R})$ on $\widehat{\mathbb{C}}$ stabilizes $\mathcal{H}$.*

1.2. Congruence Subgroups. For $N \in \mathbb{Z}_{\geq 1}$, let

$$\Gamma(N) = \left\{ g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{N} \right\}.$$

This is the *principal congruence subgroup of level* N.

Proposition 11.1.4 *Let $N \geq 1$. Then:*

(1) The reduction map

$$\pi_N : \mathrm{SL}_2(\mathbb{Z}) \longrightarrow \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$$

is surjective, with kernel $\Gamma(N)$. In particular,

$$\mathrm{SL}_2(\mathbb{Z})/\Gamma(N) \cong \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}).$$

(2) The index of $\Gamma(N)$ in $\mathrm{SL}_2(\mathbb{Z})$ is

$$[\mathrm{SL}_2(\mathbb{Z}) : \Gamma(N)] = N^3 \prod_{p \mid N} \left(1 - \frac{1}{p^2}\right).$$

Proof.

(1) Let $\gamma \in \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$. Choose an integral lift

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{Z}), \qquad A \equiv \gamma \pmod{N}.$$

Consider the set of all lifts

$$S = \left\{ A + N \begin{pmatrix} k & \ell \\ s & t \end{pmatrix} : k, \ell, s, t \in \mathbb{Z} \right\}.$$

We want to prove $S \cap \mathrm{SL}_2(\mathbb{Z}) \neq \varnothing$.

Step 1. Find c' and d' with $\gcd(c', d') = 1$. We seek integers s, t such that

$$c' = c + sN, \qquad d' = d + tN, \qquad \gcd(c', d') = 1.$$

Since $\det \gamma = 1$ in $\mathbb{Z}/N\mathbb{Z}$, we have

$$ad - bc \equiv 1 \pmod{N}.$$

Set $g = \gcd(c, d)$. As $g \mid ad - bc$, this implies

$$\boxed{\gcd(g, N) = 1.}$$

(The case $N = 1$ is immediate, so we may assume $N > 1$.)

The argument is symmetric in c and d. Moreover, c and d cannot both be zero, since $ad - bc \equiv 1 \pmod{N}$. Thus, without loss of generality, assume $c \neq 0$ and take $s = 0$. In other words, put $c' = c$ and look for

$$d' = d + tN \qquad \text{with} \qquad \gcd(c, d + tN) = 1.$$

By the Chinese remainder theorem, there exists $t \in \mathbb{Z}$ such that

$$\boxed{\begin{cases} t \equiv 1 \pmod{p}, & \text{for every prime } p \mid g, \\ t \equiv 0 \pmod{p}, & \text{for every prime } p \mid c \text{ with } p \nmid g. \end{cases}}$$

We now check that $\gcd(c, d + tN) = 1$. Let p be any prime divisor of c.

- If $p \mid g$, then $p \mid d$ but $p \nmid N$, since $\gcd(g, N) = 1$. Hence

$$d + tN \equiv N \not\equiv 0 \pmod{p}.$$

- If $p \nmid g$, then $p \nmid d$, and the choice $t \equiv 0 \pmod{p}$ gives

$$d + tN \equiv d \not\equiv 0 \pmod{p}.$$

Thus no prime divisor of c divides $d + tN$, proving

$$\boxed{\gcd(c', d') = \gcd(c, d + tN) = 1.}$$

Step 2. Find $k, \ell \in \mathbb{Z}$ to make the determinant equal to 1. We want to find integers k, ℓ such that

$$\det \begin{pmatrix} a + kN & b + \ell N \\ c' & d' \end{pmatrix} = 1.$$

Since $c' \equiv c$ and $d' \equiv d \pmod{N}$ and $ad - bc \equiv 1 \pmod{N}$, we have $ad' - bc' \equiv 1 \pmod{N}$. Thus

$$m \coloneqq \frac{1 - (ad' - bc')}{N} \in \mathbb{Z}.$$

Expanding the determinant, the required equation becomes

$$(ad' - bc') + N(kd' - \ell c') = 1, \qquad \text{equivalently} \qquad kd' - \ell c' = m.$$

By Step 1, $\gcd(c', d') = 1$. Hence Bézout's identity gives $u, v \in \mathbb{Z}$ such that $ud' + vc' = 1$. Taking $k = mu$ and $\ell = -mv$ gives $kd' - \ell c' = m$, as required. Therefore

$$\begin{pmatrix} a + kN & b + \ell N \\ c' & d' \end{pmatrix} \in S \cap \mathrm{SL}_2(\mathbb{Z}).$$

Consequently $S \cap \mathrm{SL}_2(\mathbb{Z}) \neq \varnothing$, so π_N is surjective. Its kernel is $\Gamma(N)$ by definition; the first isomorphism theorem gives

$$\mathrm{SL}_2(\mathbb{Z})/\Gamma(N) \cong \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}).$$

(2) By (1),

$$[\mathrm{SL}_2(\mathbb{Z}) : \Gamma(N)] = \#\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}).$$

Write $N = \prod_{p \mid N} p^{n_p}$. The Chinese remainder theorem gives an isomorphism of rings

$$\mathbb{Z}/N\mathbb{Z} \cong \prod_{p \mid N} \mathbb{Z}/p^{n_p}\mathbb{Z}.$$

which induces a group isomorphism

$$\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}) \cong \prod_{p|N} \mathrm{SL}_2(\mathbb{Z}/p^{n_p}\mathbb{Z}).$$

It therefore suffices to compute $\#\mathrm{SL}_2(\mathbb{Z}/p^e\mathbb{Z})$ for a prime p and $e \geq 1$.

Case 1 $e > 1$. For $e \geq 2$, consider the reduction homomorphism

$$\lambda_e : \mathrm{SL}_2(\mathbb{Z}/p^e\mathbb{Z}) \longrightarrow \mathrm{SL}_2(\mathbb{Z}/p^{e-1}\mathbb{Z}).$$

This map is surjective by (1): the surjective reduction $\pi_{p^{e-1}}$ factors as $\lambda_e \circ \pi_{p^e}$. An element lies in $\ker \lambda_e$ precisely when it has the form

$$I + p^{e-1}A \pmod{p^e}, \qquad A \in M_2(\mathbb{F}_p),$$

where any integral lift of A may be used. As $2(e-1) \geq e$,

$$\det(I + p^{e-1}A) \equiv 1 + p^{e-1}\operatorname{tr}(A) \pmod{p^e}.$$

Thus the determinant is 1 modulo p^e exactly when $\operatorname{tr}(A) = 0$ in $\mathbb{F}_p$. Since three entries of A can be chosen freely and the fourth is fixed by the trace condition,

$$\#\ker \lambda_e = p^3.$$

It follows that

$$\#\mathrm{SL}_2(\mathbb{Z}/p^e\mathbb{Z}) = p^3 \#\mathrm{SL}_2(\mathbb{Z}/p^{e-1}\mathbb{Z}) = p^{3(e-1)} \#\mathrm{SL}_2(\mathbb{F}_p).$$

Case 2 $e = 1$. We use the determinant homomorphism:

$$1 \longrightarrow \mathrm{SL}_2(\mathbb{F}_p) \longrightarrow \mathrm{GL}_2(\mathbb{F}_p) \xrightarrow{\det} \mathbb{F}_p^\times \longrightarrow 1.$$

It is surjective. To count $\mathrm{GL}_2(\mathbb{F}_p)$, choose a nonzero first column ($p^2 - 1$ choices) and then a second column outside its span ($p^2 - p$ choices). Thus

$$\#\mathrm{GL}_2(\mathbb{F}_p) = (p^2-1)(p^2-p), \qquad \#\mathbb{F}_p^\times = p - 1.$$

Hence

$$\#\mathrm{SL}_2(\mathbb{F}_p) = \frac{(p^2-1)(p^2-p)}{p-1} = p(p^2-1) = p^3\left(1 - \frac{1}{p^2}\right).$$

Consequently,

$$\#\mathrm{SL}_2(\mathbb{Z}/p^e\mathbb{Z}) = p^{3e}\left(1 - \frac{1}{p^2}\right).$$

Taking the product over all primes dividing N proves

$$[\mathrm{SL}_2(\mathbb{Z}) : \Gamma(N)] = \prod_{p|N} p^{3n_p}\left(1 - \frac{1}{p^2}\right) = \boxed{N^3 \prod_{p|N}\left(1 - \frac{1}{p^2}\right)}.$$

□

Definition 11.1.5 A subgroup Γ of $\mathrm{SL}_2(\mathbb{Z})$ is a *congruence subgroup* if $\Gamma(N) \subseteq \Gamma$ for some N. ⌟

Example 11.1.6

$$\Gamma_1(N) = \left\{ g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \pmod{N} \right\},$$

$$\Gamma_0(N) = \left\{ g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \pmod{N} \right\}.$$

Here $*$ means no condition on the entry.

$$\Gamma(N) \subseteq \Gamma_1(N) \subseteq \Gamma_0(N) \subseteq \mathrm{SL}_2(\mathbb{Z}).$$

Proposition 11.1.7 *Let $N \in \mathbb{N}_{\geq 1}$.*

1) The map

$$\lambda_1 : \Gamma_1(N) \longrightarrow \mathbb{Z}/N\mathbb{Z}, \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \longmapsto b \pmod{N}$$

is a surjective group homomorphism with kernel $\Gamma(N)$.

2) The map

$$\lambda_0 : \Gamma_0(N) \longrightarrow (\mathbb{Z}/N\mathbb{Z})^\times, \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \longmapsto d \pmod{N}$$

is a surjective group homomorphism with kernel $\Gamma_1(N)$.

Indeed,

$$\lambda_1(gg') = \lambda_1(g) + \lambda_1(g') \pmod{N}.$$

To see that λ_0 is well-defined, note that

$$1 = \det(g) = ad - bc \equiv ad \pmod{N} \quad \Longrightarrow \quad d \in (\mathbb{Z}/N\mathbb{Z})^\times.$$

Also,

$$\lambda_0(gg') = \lambda_0(g)\lambda_0(g').$$

Summary

$\mathrm{SL}_2(\mathbb{Z})$ acts on $\mathcal{H}$.

$$0 \longrightarrow \Gamma(N) \longrightarrow \mathrm{SL}_2(\mathbb{Z}) \longrightarrow \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}) \longrightarrow 0,$$

$$0 \longrightarrow \Gamma(N) \longrightarrow \Gamma_1(N) \longrightarrow \mathbb{Z}/N\mathbb{Z} \longrightarrow 0,$$

$$0 \longrightarrow \Gamma_1(N) \longrightarrow \Gamma_0(N) \longrightarrow (\mathbb{Z}/N\mathbb{Z})^\times \longrightarrow 0.$$

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & ax + b \\ 0 & d \end{pmatrix}.$$

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \begin{pmatrix} 1 & -d^{-1}b \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}.$$

1.3. Weak Modular Functions.

Definition 11.1.8 For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ and $z \in \mathcal{H}$, the *factor of automorphy* is

$$j(g, z) = cz + d \in \mathbb{C}^\times.$$

⌟

Definition 11.1.9 For $g \in \mathrm{SL}_2(\mathbb{Z})$ and $k \in \mathbb{Z}$, the weight-k operator $[g]_k$ associates to a function $f : \mathcal{H} \to \mathbb{C}$ the function $f[g]_k : \mathcal{H} \to \mathbb{C}$ defined by

$$(f[g]_k)(z) = j(g, z)^{-k} f(gz), \qquad z \in \mathcal{H}.$$

⌟

Lemma 11.1.10 *Let $g, g' \in \mathrm{SL}_2(\mathbb{Z})$, $z \in \mathcal{H}$, and $f : \mathcal{H} \to \mathbb{C}$. For every $k \in \mathbb{Z}$, the following identities hold:*

(1) $j(gg', z) = j(g, g'z)\, j(g', z)$.

(2) $f[gg']_k = (f[g]_k)[g']_k$.

(3) $\dfrac{d}{dz}(gz) = \dfrac{1}{j(g, z)^2}$.

These three identities can be verified by direct calculation and are left as an exercise.

Definition 11.1.11 Let Γ be a congruence subgroup and let $k \in \mathbb{Z}$. A function $f : \mathcal{H} \to \mathbb{C}$ is *weakly modular of weight k* if

(1) f is meromorphic;

(2) $f[\gamma]_k = f$ for every $\gamma \in \Gamma$.

⌟

Note: By Lemma 11.1.2, $j(g, z)$ is nonzero on the upper half-plane. Hence if f is meromorphic, then $f[g]_k$ is also meromorphic.

Remark 11.1.12 When $k = 0$, the weak modularity condition is $f(\gamma z) = f(z)$ for every $\gamma \in \Gamma$. In other words, f is a Γ-invariant meromorphic function. In particular, constant functions are weakly modular of weight 0. ⌟

Remark 11.1.13 If $-I = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \in \Gamma$ and f is weakly modular of weight k with respect to Γ, then

$$f(z) = (f[-I]_k)(z) = (-1)^{-k} f(z).$$

Thus if k is odd, then $f \equiv 0$. ⌟

Remark 11.1.14 By the definition of a congruence subgroup, there exists $N \geqslant 1$ such that $\Gamma(N) \subseteq \Gamma$. In particular,

$$\begin{pmatrix} 1 & N \\ 0 & 1 \end{pmatrix} \in \Gamma.$$

Hence there exists a least positive integer h such that

$$\gamma_h = \begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix} \in \Gamma.$$

If f is weakly modular of weight k with respect to Γ, then

$$f(z) = (f[\gamma_h]_k)(z) = (0z+1)^{-k} f(\gamma_h z) = f(z+h).$$

Thus f is $h\mathbb{Z}$-periodic. We will use this when introducing q-expansions. ⌟

Remark 11.1.15 Suppose that the congruence subgroup $\Gamma = \langle \gamma_1, \cdots, \gamma_n \rangle$.

By Lemma 11.1.10(2), to check weak modularity for a meromorphic function f, it suffices to verify

$$f[\gamma_i]_k = f, \qquad i = 1, \cdots, n,$$

These conditions imply that f is weakly modular of weight k with respect to Γ. Indeed, every element of Γ is a product of generators and their inverses, and the composition law shows that invariance under the generators implies invariance under their inverses and all such products. ⌟

Example 11.1.16 $\mathrm{SL}_2(\mathbb{Z})$ is generated by the following two matrices:

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Hence a meromorphic function $f : \mathcal{H} \to \mathbb{C}$ satisfying $f[S]_k = f$ and $f[T]_k = f$ is weakly modular of weight k with respect to $\mathrm{SL}_2(\mathbb{Z})$. Explicitly, these two conditions are

$$f(z) = z^{-k} f(-1/z), \qquad f(z+1) = f(z) (\mathbb{Z}\text{-periodic}).$$

Lemma 11.1.17

(i) If Γ is a congruence subgroup and $\alpha \in \mathrm{SL}_2(\mathbb{Z})$, then $\alpha^{-1}\Gamma\alpha$ is a congruence subgroup.

(ii) If f is weakly modular of weight k with respect to Γ and $\alpha \in \mathrm{SL}_2(\mathbb{Z})$, then $f[\alpha]_k$ is weakly modular of weight k with respect to $\alpha^{-1}\Gamma\alpha$.

Proof.

(i) Let $\Gamma(N) \subseteq \Gamma$. Since $\Gamma(N) \trianglelefteq \mathrm{SL}_2(\mathbb{Z})$,

$$\Gamma(N) = \alpha^{-1}\Gamma(N)\alpha \subseteq \alpha^{-1}\Gamma\alpha.$$

(ii) By definition and Lemma 11.1.10(2),

$$(f[\alpha]_k)[\alpha^{-1}\gamma\alpha]_k = f[\alpha\alpha^{-1}\gamma\alpha]_k = f[\gamma\alpha]_k = (f[\gamma]_k)[\alpha]_k = f[\alpha]_k$$

for $\gamma \in \Gamma$.

□

1.4. Cusps. Put $X = \mathbb{Q} \cup \{\infty\} = \mathbb{P}^1(\mathbb{Q})$. Since $\mathrm{SL}_2(\mathbb{Z})$ stabilizes X, its action on X is well-defined. We call the Γ-orbits in X the *cusps* of Γ.

Let Γ be a congruence subgroup. For $x \in X$, its Γ-*orbit* is

$$\Gamma \cdot x = \{\gamma x : \gamma \in \Gamma\}.$$

We write $x \sim_\Gamma y$ if $y = \gamma x$ for some $\gamma \in \Gamma$, and denote the corresponding orbit (equivalence class) by $[x]_\Gamma = \Gamma \cdot x$.

Definition 11.1.18 The *set of cusps* of Γ is the orbit set

$$\Gamma\backslash X = \Gamma\backslash\big(\mathbb{Q} \cup \{\infty\}\big) = \{[x]_\Gamma : x \in X\}.$$

⌟

Lemma 11.1.19

(i) $\mathrm{SL}_2(\mathbb{Z})\backslash X = \{[\infty]_{\mathrm{SL}_2(\mathbb{Z})}\}$.

(ii) The set of cusps $\Gamma\backslash X$ *is finite.*

Proof.

(i) Let $x = a/c \in \mathbb{Q}$, with $a, c \in \mathbb{Z}$ and $\gcd(a, c) = 1$. By Bézout's identity there are $b, d \in \mathbb{Z}$ such that $ad - bc = 1$. Hence

$$\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}), \qquad \gamma\infty = \frac{a}{c} = x.$$

The point ∞ itself is in its own orbit, so every element of X is in the orbit of ∞.

(ii) Put $G = \mathrm{SL}_2(\mathbb{Z})$. Since Γ is a congruence subgroup, there exists $N \geq 1$ with $H \coloneqq \Gamma(N) \subseteq \Gamma$. By Proposition 11.1.4, $H \trianglelefteq G$ and G/H is finite. Write $\overline{g} = gH \in G/H$ and $[x]_H = H \cdot x \in H\backslash X$. The group G/H acts on the H-orbit space $H\backslash X$ via

$$\overline{g} \cdot [x]_H \coloneqq [gx]_H.$$

This is well-defined because H is normal in G. Taking orbits in two stages gives

$$(G/H)\backslash(H\backslash X) \cong G\backslash X = \{[\infty]_G\}.$$

Thus the finite group G/H acts transitively on $H\backslash X$; in particular,

$$|H\backslash X| \leq |G/H| < \infty.$$

Since $H \subseteq \Gamma$, there is a natural surjection $H\backslash X \twoheadrightarrow \Gamma\backslash X$, so

$$|\Gamma\backslash X| < \infty.$$

□

Let

$$D = \{q \in \mathbb{C} : |q| < 1\}, \qquad D' = D \setminus \{0\}$$

be the *open complex unit disk* and the *punctured open complex unit disk*, respectively. Let $h \in \mathbb{N}_{>0}$. The function

$$q_h : \mathcal{H} \longrightarrow D', \qquad z \longmapsto q_h(z) = e^{2\pi i z/h}$$

is $h\mathbb{Z}$-periodic.

Let $f : \mathcal{H} \to \mathbb{C}$ be an $h\mathbb{Z}$-periodic function. Since q_h identifies $\mathcal{H}/h\mathbb{Z}$ with D', the function f factors through q_h: there is a function $\widetilde{f} : D' \to \mathbb{C}$ such that

$$f(z) = \widetilde{f}(q_h(z)).$$

Definition 11.1.20 We say that f is *meromorphic* (respectively, *holomorphic*) *at* ∞ if $\widetilde{f}$ extends meromorphically (respectively, holomorphically) across $0 \in D$. ⌟

If f is meromorphic at ∞, then $\widetilde{f}$ has a Laurent expansion at 0. Thus f has a *Fourier expansion* (or q_h*-expansion*)

$$\boxed{f(z) = \sum_{n=-m}^{\infty} a(n)\, q_h(z)^n, \qquad m \geq 0,}$$

valid for $\mathrm{Im}(z)$ sufficiently large. In the holomorphic case there are no negative powers (we can take $m = 0$), and

$$f(\infty) := \widetilde{f}(0) = a(0).$$

1.5. Modular Forms.

Definition 11.1.21 Let $k \in \mathbb{Z}$ and let Γ be a congruence subgroup. A *modular form of weight* k with respect to Γ is a function $f : \mathcal{H} \to \mathbb{C}$ such that

(1) f is holomorphic on $\mathcal{H}$;

(2) f is weakly modular of weight k with respect to Γ;

(3) for every $\alpha \in \mathrm{SL}_2(\mathbb{Z})$, the function $f[\alpha]_k$ is holomorphic at ∞.

⌟

Definition 11.1.22 A modular form of weight k with respect to Γ is a *cusp form* if

$$\forall \alpha \in \mathrm{SL}_2(\mathbb{Z}), \qquad (f[\alpha]_k)(\infty) = 0.$$

⌟

Remark 11.1.23 Condition (3) of Definition 11.1.21 only needs to be checked for finitely many α. Indeed, the congruence subgroup Γ has finite index. Choose representatives $\alpha_1, \cdots, \alpha_r$ of the left-coset space $\Gamma\backslash\mathrm{SL}_2(\mathbb{Z})$. Every $\alpha \in \mathrm{SL}_2(\mathbb{Z})$ can be written as $\alpha = \gamma\alpha_i$ with $\gamma \in \Gamma$. By Lemma 11.1.10(2),

$$f[\alpha]_k = f[\gamma\alpha_i]_k = (f[\gamma]_k)[\alpha_i]_k = f[\alpha_i]_k.$$

Thus it suffices to check that each $f[\alpha_i]_k$ is holomorphic at ∞. ⌟

Example 11.1.24 Equipped with the usual sum $f+f'$ and scalar multiplication uf ($u \in \mathbb{C}$), the set of modular forms of weight k with respect to Γ forms a $\mathbb{C}$-vector space, denoted

$$\mathcal{M}_k(\Gamma).$$

The subset of cusp forms is a $\mathbb{C}$-vector subspace of $\mathcal{M}_k(\Gamma)$, denoted

$$\mathcal{S}_k(\Gamma).$$

Moreover, the usual pointwise product of functions induces a graded-ring structure on

$$\mathcal{M}(\Gamma) \coloneqq \bigoplus_{k\in\mathbb{Z}} \mathcal{M}_k(\Gamma),$$

and

$$\mathcal{S}(\Gamma) \coloneqq \bigoplus_{k\in\mathbb{Z}} \mathcal{S}_k(\Gamma)$$

defines an ideal of $\mathcal{M}(\Gamma)$.

1.6. Eisenstein Series.

Definition 11.1.25 Let $k > 2$ be an even integer. The *Eisenstein series of weight* k is the function

$$G_k : \mathcal{H} \to \mathbb{C}, \quad z \mapsto \sum_{(c,d)\in\mathbb{Z}^2\setminus\{(0,0)\}} \frac{1}{(cz+d)^k}.$$

We write $\sum_{(c,d)}{}'$ for a sum over all $(c,d) \in \mathbb{Z}^2 \setminus \{(0,0)\}$. ⌟

Proposition 11.1.26 *Let $k > 2$ be an even integer. Then $G_k \in \mathcal{M}_k(\mathrm{SL}_2(\mathbb{Z}))$.*

Proof.

(1) G_k is holomorphic on $\mathcal{H}$. We prove absolute convergence on $\mathcal{H}$ and uniform convergence on each region

$$\Omega(A,B) \coloneqq \{z = x+iy \in \mathcal{H} \mid |x| \leqslant A,\ y \geqslant B\}, \quad A, B > 0.$$

We seek a constant $K > 0$, depending only on A and B, such that

$$|cz+d|^2 \geqslant K|ci+d|^2 = K(c^2+d^2) \tag{11.1.1}$$

for every $z \in \Omega(A,B)$ and $(c,d) \neq (0,0)$. Take

$$K = \frac{B^2}{1+(A+B)^2} < 1.$$

If $c = 0$, (11.1.1) follows from $K < 1$. If $c \neq 0$, put $\lambda = d/c$. It suffices to show that

$$\frac{(\lambda+x)^2+y^2}{1+\lambda^2} \geqslant K.$$

Case 1 $|\lambda| \leqslant A+B$. Since $y \geqslant B$,

$$\frac{(\lambda+x)^2+y^2}{1+\lambda^2} \geqslant \frac{B^2}{1+(A+B)^2} = K.$$

Case 2 $|\lambda| > A + B$. We have

$$\left|\frac{x}{\lambda}\right| \leqslant \frac{A}{|\lambda|} < \frac{A}{A+B} < 1.$$

By the reverse triangle inequality,

$$\left|1 + \frac{x}{\lambda}\right| \geqslant 1 - \left|\frac{x}{\lambda}\right| > \frac{B}{A+B}.$$

Hence

$$\frac{(\lambda+x)^2 + y^2}{1+\lambda^2} \geqslant \frac{B^2}{(A+B)^2}\frac{\lambda^2}{1+\lambda^2}.$$

The function $t \mapsto t/(1+t)$ is increasing for $t \geqslant 0$, so

$$\frac{\lambda^2}{1+\lambda^2} \geqslant \frac{(A+B)^2}{1+(A+B)^2}.$$

Combining these inequalities gives the required lower bound K.

By (11.1.1),

$$\frac{1}{|cz+d|^k} \leqslant \frac{K^{-k/2}}{|ci+d|^k}.$$

The series

$$\sum_{(c,d)}{}' \frac{1}{|ci+d|^k} = \sum_{(c,d)}{}' \frac{1}{(c^2+d^2)^{k/2}}$$

converges for $k > 2$; this is left as an exercise. The Weierstrass M-test gives absolute and uniform convergence on $\Omega(A,B)$. Every compact subset of $\mathcal{H}$ is contained in some $\Omega(A,B)$, and each summand is holomorphic by Lemma 11.1.2. Therefore G_k is holomorphic on $\mathcal{H}$.

(2) G_k is weakly modular of weight k with respect to $\mathrm{SL}_2(\mathbb{Z})$. Let $g = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$. Then

$$\begin{aligned} G_k(gz) &= \sum_{(c,d)}{}' \frac{1}{\left(c\frac{\alpha z+\beta}{\gamma z+\delta} + d\right)^k} \\ &= (\gamma z+\delta)^k \sum_{(c,d)}{}' \frac{1}{((c\alpha+d\gamma)z + (c\beta+d\delta))^k}. \end{aligned}$$

The map

$$\mathbb{Z}^2 \setminus \{(0,0)\} \to \mathbb{Z}^2 \setminus \{(0,0)\}, \quad (c,d) \mapsto (c,d)g = (c\alpha+d\gamma, c\beta+d\delta)$$

is a bijection with inverse $(c,d) \mapsto (c,d)g^{-1}$. By absolute convergence, we may reindex the sum, obtaining

$$G_k(gz) = (\gamma z+\delta)^k G_k(z),$$

i.e. $G_k[g]_k = G_k$.

(3) G_k is holomorphic at ∞. By (2), $G_k(z+1) = G_k(z)$. Hence G_k induces a holomorphic function $\widetilde{G}_k$ on the punctured unit disk through $q = e^{2\pi i z}$. It suffices to show that $\widetilde{G}_k$ is bounded near $q = 0$.

Fix $B > 0$. By (1), G_k is bounded on $\Omega(1, B)$ by the finite constant

$$K^{-k/2} \sum_{(c,d)}{}' \frac{1}{|ci + d|^k}.$$

By periodicity, every z with $\mathrm{Im}(z) \geqslant B$ can be translated into $\Omega(1, B)$ without changing $G_k(z)$. Thus $\widetilde{G}_k$ is bounded for $0 < |q| \leqslant e^{-2\pi B}$ and extends holomorphically across $q = 0$ by the removable singularity theorem. Finally, $G_k[\alpha]_k = G_k$ for every $\alpha \in \mathrm{SL}_2(\mathbb{Z})$, so all the conditions in Definition 11.1.21 hold.

□

Proposition 11.1.27 (Fourier Expansion) *Let $k > 2$ be an even integer and $q = e^{2\pi i z}$. Then*

$$G_k(z) = 2\zeta(k) + \frac{2(2\pi i)^k}{(k-1)!} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n,$$

where

$$\zeta(k) = \sum_{m=1}^{\infty} \frac{1}{m^k}, \qquad \sigma_{k-1}(n) = \sum_{\substack{m \mid n \\ m>0}} m^{k-1}.$$

Proof.

Step 1. We use the identities

$$\pi \cot(\pi z) = \pi \frac{\cos(\pi z)}{\sin(\pi z)} = \frac{1}{z} + \sum_{m=1}^{\infty} \left(\frac{1}{z+m} + \frac{1}{z-m} \right).$$

The partial fraction expansion is quoted here; its justification is deferred to a later lecture. For $z \in \mathcal{H}$, $|q| < 1$, and the exponential formulas for sine and cosine give

$$\begin{aligned} \pi \cot(\pi z) &= i\pi \frac{q+1}{q-1} = -i\pi \frac{1+q}{1-q} \\ &= -i\pi(1+q) \sum_{n=0}^{\infty} q^n = i\pi - 2\pi i \sum_{n=0}^{\infty} q^n. \end{aligned}$$

Differentiating $k-1$ times yields

$$\sum_{m \in \mathbb{Z}} \frac{1}{(m+z)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{n=1}^{\infty} n^{k-1} q^n. \tag{11.1.2}$$

Step 2. Separate the terms with $c = 0$ and $c \neq 0$. Since k is even,

$$\begin{aligned} G_k(z) &= \sum_{d \neq 0} \frac{1}{d^k} + \sum_{c \neq 0} \sum_{d \in \mathbb{Z}} \frac{1}{(cz+d)^k} \\ &= 2\zeta(k) + 2 \sum_{c=1}^{\infty} \sum_{d \in \mathbb{Z}} \frac{1}{(cz+d)^k}. \end{aligned}$$

By (11.1.2) applied to cz,

$$2 \sum_{c=1}^{\infty} \sum_{d \in \mathbb{Z}} \frac{1}{(cz+d)^k} = \frac{2(2\pi i)^k}{(k-1)!} \sum_{c=1}^{\infty} \sum_{m=1}^{\infty} m^{k-1} q^{cm}.$$

Grouping equal powers of q gives

$$\sum_{c=1}^{\infty}\sum_{m=1}^{\infty} m^{k-1}q^{cm} = \sum_{n=1}^{\infty}\left(\sum_{\substack{m|n\\ m>0}} m^{k-1}\right)q^n = \sum_{n=1}^{\infty}\sigma_{k-1}(n)q^n.$$

This proves the formula.

□

1.7. The Discriminant and the Modular Invariant.

Definition 11.1.28 Define

$$g_2(z) \coloneqq 60G_4(z), \qquad g_3(z) \coloneqq 140G_6(z).$$

The *discriminant function* is

$$\Delta : \mathcal{H} \to \mathbb{C}, \quad z \mapsto g_2(z)^3 - 27g_3(z)^2.$$

⌟

Since $G_4 \in \mathcal{M}_4(\mathrm{SL}_2(\mathbb{Z}))$, $G_6 \in \mathcal{M}_6(\mathrm{SL}_2(\mathbb{Z}))$ and $\mathcal{M}(\mathrm{SL}_2(\mathbb{Z}))$ is a graded ring, we have

$$\Delta \in \mathcal{M}_{12}(\mathrm{SL}_2(\mathbb{Z})).$$

Note: Using the expression for $\zeta(2k)$ in terms of Bernoulli numbers, verify that $\Delta(\infty) = 0$. Thus $\Delta \in \mathcal{S}_{12}(\mathrm{SL}_2(\mathbb{Z}))$. This is left as an exercise.

Remark 11.1.29 We will later prove that $\Delta(z) \neq 0$ for every $z \in \mathcal{H}$. In particular, Δ is a nonzero cusp form of weight 12. ⌟

Definition 11.1.30 The *modular invariant* is the function

$$j : \mathcal{H} \to \mathbb{C}, \quad z \mapsto 1728\frac{g_2(z)^3}{\Delta(z)}.$$

⌟

Remark 11.1.31 Using the nonvanishing of Δ on $\mathcal{H}$, the properties of G_4 and Δ show that j is holomorphic on $\mathcal{H}$ and invariant under $\mathrm{SL}_2(\mathbb{Z})$. We will also see later that j has a simple pole at ∞. ⌟

Note: A goal of the course is to compute $\dim \mathcal{M}_k(\Gamma)$ and relate modular forms to the geometry of the modular curve $X(\Gamma)$, complex tori and elliptic curves.

2. Modular Curves

2.1. Riemann Surfaces.

Definition 11.2.1 A *complete one-dimensional complex atlas* on a Hausdorff topological space X is a collection $\{(U_i, \varphi_i)\}_{i\in I}$ satisfying:

(1) $\{U_i\}_{i\in I}$ is an open cover of X.

(2) Each $\varphi_i : U_i \to \mathbb{C}$ is a homeomorphism onto its image.

(3) For every $i, j \in I$, the transition map

$$\varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \to \varphi_j(U_i \cap U_j)$$

is holomorphic.

(4) If $V \subseteq X$ is open and $\psi : V \to \mathbb{C}$ is a homeomorphism onto its image such that $\psi \circ \varphi_i^{-1}$ is holomorphic on $\varphi_i(U_i \cap V)$ for every $i \in I$, then (V, ψ) belongs to the atlas.

⌟

Definition 11.2.2 A *Riemann surface* is a connected Hausdorff topological space equipped with a complete one-dimensional complex atlas. It is *compact* if its underlying topological space is compact. ⌟

Example 11.2.3

(1) Noncompact examples include $\mathbb{C}$, connected subsets of $\mathbb{C}$ and modular curves $Y(\Gamma)$.

(2) Compact examples include the Riemann sphere $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$, complex tori $\mathbb{C}/\Lambda$, modular curves $X(\Gamma)$ and g-holed tori.

Definition 11.2.4 A map $f : X \to Y$ between Riemann surfaces is *holomorphic* if it is continuous and, for every chart (U, φ) on X and (V, ψ) on Y with $f(U) \subseteq Y$, the coordinate expression

$$\psi \circ f \circ \varphi^{-1} : \varphi(U) \to \psi(V)$$

is holomorphic. ⌟

Proposition 11.2.5 (Local Normal Form) *Let $f : X \to Y$ be a nonconstant holomorphic map between Riemann surfaces, $x \in X$ and $y = f(x)$. There exist an integer $e_x > 0$ and charts (U, φ) on X and (V, ψ) on Y such that:*

(1) $x \in U$, $\varphi(x) = 0$, $y \in V$ and $\psi(y) = 0$.

(2) $f(U) \subseteq V$.

(3) In these coordinates,

$$\psi \circ f \circ \varphi^{-1} : \varphi(U) \to \psi(V), \quad z \mapsto z^{e_x}.$$

Sketch of proof. Choose charts satisfying (1) and (2), and put $h = \psi \circ f \circ \varphi^{-1}$. By the identity theorem, h is nonconstant. Since $h(0) = 0$, write

$$h(z) = z^m g(z), \quad m > 0, \quad g(0) \neq 0,$$

where g is holomorphic. After shrinking the charts, choose a holomorphic m-th root of g. The map

$$t(z) = z \sqrt[m]{g(z)}$$

fixes 0 and has nonzero derivative $t'(0) = \sqrt[m]{g(0)}$. Shrinking the charts again, t is biholomorphic. Replacing φ by $t \circ \varphi$, the coordinate expression of f becomes $z \mapsto z^m$. □

Definition 11.2.6 Let $f : X \to Y$ be a nonconstant holomorphic map. The integer e_x in Proposition 11.2.5 is the *ramification index* at x. A point x is a *ramification point* if $e_x > 1$. Write

$$R(f) \coloneqq \{x \in X \mid e_x > 1\}.$$

⌟

Corollary 11.2.7 *Let $f : X \to Y$ be a nonconstant holomorphic map between Riemann surfaces.*

(1) For every $y \in Y$, the fibre $f^{-1}(y)$ is a closed discrete subset of X.

(2) $R(f)$ is a closed discrete subset of X.

In particular, if X is compact, then $f^{-1}(y)$ and $R(f)$ are finite.

Proof. Locally, f has the form $z \mapsto z^m$, which has discrete fibres and at most one ramification point, namely 0 when $m > 1$. The fibre is closed by continuity, and the local normal form also shows that the complement of $R(f)$ is open. Closed discrete subsets of a compact space are finite. □

Corollary 11.2.8 *A nonconstant holomorphic map between Riemann surfaces is open.*

Proof. Locally, the map has the form $z \mapsto z^m$, which is open. □

Corollary 11.2.9 *Let $f : X \to Y$ be a holomorphic map between Riemann surfaces. If X is compact, then f is either constant or surjective.*

Proof. Assume f is nonconstant.

- $f(X)$ is open in Y by Corollary 11.2.8.
- $f(X)$ is compact. Since Y is Hausdorff, $f(X)$ is closed in Y.

Since Y is connected, $f(X) = Y$. □

Theorem-Definition 11.2.10 (Degree) *Let $f : X \to Y$ be a holomorphic map between Riemann surfaces with X compact. The degree of f is the integer*

$$\deg(f) \coloneqq \sum_{x \in f^{-1}(y)} e_x,$$

independent of the choice of $y \in Y$. ⌟

Sketch of proof. One needs to check independence of y. Away from the branch values, use the covering map supplied by the local normal form. An evenly covered neighbourhood has inverse image a disjoint union of open sets, each mapped homeomorphically onto that neighbourhood. Near a ramification point, the local expression $z \mapsto z^{e_x}$ accounts for the multiplicity e_x. □

2.2. Complex Tori.

Definition 11.2.11 A *lattice* in $\mathbb{C}$ is a subgroup

$$\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2 \subseteq \mathbb{C},$$

where $\{\omega_1, \omega_2\}$ is a basis of $\mathbb{C}$ over $\mathbb{R}$. ⌟

We adopt the convention that $\omega_1/\omega_2 \in \mathcal{H}$.

Lemma 11.2.12 *Let $\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$ and $\Lambda' = \mathbb{Z}\omega_1' \oplus \mathbb{Z}\omega_2'$ be lattices with bases satisfying the above convention. Then $\Lambda = \Lambda'$ if and only if there exists $g \in \mathrm{SL}_2(\mathbb{Z})$ such that*

$$\begin{pmatrix} \omega_1' \\ \omega_2' \end{pmatrix} = g \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix}.$$

Proof.

(1) Suppose $\Lambda = \Lambda'$. There exists $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{Z})$ taking one basis to the other. Put $z = \omega_1/\omega_2$ and $z' = \omega_1'/\omega_2'$. Then

$$gz = \frac{az+b}{cz+c} = \frac{a\omega_1 + b\omega_2}{c\omega_1 + d\omega_2} = \frac{\omega_1'}{\omega_2'} = z'.$$

Since $z, z' \in \mathcal{H}$, Proposition 11.1.1 implies $\det(g) > 0$. As $\det(g) = \pm 1$, we have $\det(g) = 1$.

(2) Conversely, a matrix in $\mathrm{SL}_2(\mathbb{Z})$ is an invertible integral change of basis, so the two bases generate the same lattice.

□

Definition 11.2.13 A *complex torus* is a quotient of $\mathbb{C}$ by a lattice Λ:

$$\mathbb{C}/\Lambda = \{z + \Lambda \mid z \in \mathbb{C}\}.$$

⌟

Remark 11.2.14

- A complex torus is an abelian group, as a quotient of $(\mathbb{C}, +)$ by Λ.
- A complex torus is a compact Riemann surface.

In particular, a holomorphic map $\mathbb{C}/\Lambda \to \mathbb{C}/\Lambda'$ is either constant or surjective by Corollary 11.2.9. ⌟

Proposition 11.2.15 *Let $f : \mathbb{C}/\Lambda \to \mathbb{C}/\Lambda'$ be holomorphic. There exist $m, b \in \mathbb{C}$ such that:*

(1) $m\Lambda \subseteq \Lambda'$.

(2) $f(z + \Lambda) = mz + b + \Lambda'$.

Moreover, f is invertible if and only if $m\Lambda = \Lambda'$.

Proof.

Step 1. Since $\mathbb{C}$ is a universal cover of $\mathbb{C}/\Lambda'$, the composite $\mathbb{C} \to \mathbb{C}/\Lambda \xrightarrow{f} \mathbb{C}/\Lambda'$ admits a lift $\widetilde{f} : \mathbb{C} \to \mathbb{C}$. The covering-space justification is deferred to a later reference. The lift is holomorphic because f is holomorphic.

Step 2. Fix $\lambda \in \Lambda$ and define

$$g_\lambda(z) \coloneqq \widetilde{f}(z+\lambda) - \widetilde{f}(z).$$

Since $\widetilde{f}$ is a lift, g_λ takes values in Λ'. It is continuous, $\mathbb{C}$ is connected and Λ' is discrete, so g_λ is constant. Differentiating gives

$$\widetilde{f}'(z+\lambda) = \widetilde{f}'(z).$$

Thus $\widetilde{f}'$ is entire and Λ-periodic, hence bounded on $\mathbb{C}$. By Liouville's theorem it is constant. Therefore

$$\widetilde{f}(z) = mz + b.$$

The lifting property gives $m\Lambda \subseteq \Lambda'$ and $f(z+\Lambda) = mz + b + \Lambda'$.

Step 3. For the invertibility assertion, the board argues that if $m\Lambda \neq \Lambda'$, one can choose $z \in \Lambda'$ with $z/m \notin \Lambda$. Then

$$f(z/m + \Lambda) = b + \Lambda' = f(0+\Lambda),$$

so f is not injective. If $m\Lambda = \Lambda'$, the inverse is induced by an affine map on $\mathbb{C}$.

□

Proposition 11.2.16 *Let $f : \mathbb{C}/\Lambda \to \mathbb{C}/\Lambda'$ be the holomorphic map $f(z+\Lambda) = mz + b + \Lambda'$ with $m\Lambda \subseteq \Lambda'$. The following are equivalent:*

(1) f is a group homomorphism.

(2) $b \in \Lambda'$, so $f(z+\Lambda) = mz + \Lambda'$.

(3) $f(0) = 0$.

Proof.

(1) (1) $\implies$ (3) is immediate.

(2) (3) $\implies$ (2): $f(0+\Lambda) = b + \Lambda'$, so $f(0) = 0$ if and only if $b \in \Lambda'$.

(3) (2) $\implies$ (1): the map is $z + \Lambda \mapsto mz + \Lambda'$, which is a group homomorphism.

□

Corollary 11.2.17

(1) There exists a nonzero holomorphic group homomorphism $\mathbb{C}/\Lambda \to \mathbb{C}/\Lambda'$ if and only if there exists $m \in \mathbb{C}^\times$ such that $m\Lambda \subseteq \Lambda'$.

(2) *There exists a holomorphic group isomorphism* $\mathbb{C}/\Lambda \to \mathbb{C}/\Lambda'$ *if and only if there exists* $m \in \mathbb{C}^\times$ *such that* $m\Lambda = \Lambda'$.

Proof. Apply Proposition 11.2.15 and Proposition 11.2.16. □

Definition 11.2.18 A nonzero holomorphic group homomorphism between complex tori is an *isogeny*. ⌟

Remark 11.2.19 If $f : \mathbb{C}/\Lambda \to \mathbb{C}/\Lambda'$ is the isogeny $f(z+\Lambda) = mz + \Lambda'$, then

$$\ker(f) = m^{-1}\Lambda'/\Lambda = f^{-1}(\{0\})$$

is finite by Corollary 11.2.7. ⌟

Example 11.2.20 (Multiplication by an Integer) Let $m \in \mathbb{N}_{>0}$ and let Λ be a lattice. Define

$$[m] : \mathbb{C}/\Lambda \to \mathbb{C}/\Lambda, \quad z + \Lambda \mapsto mz + \Lambda.$$

Since $m\Lambda \subseteq \Lambda$, this is an isogeny. Moreover,

$$\ker([m]) = m^{-1}\Lambda/\Lambda \cong (\mathbb{Z}/m\mathbb{Z})^2.$$

If $E = \mathbb{C}/\Lambda$, we write

$$E[m] \coloneqq \ker([m] : E \to E)$$

for its *m-torsion subgroup*.

Example 11.2.21 (Quotient by a Cyclic Subgroup) Let $E = \mathbb{C}/\Lambda$, $m \in \mathbb{N}_{>0}$ and let $C \subseteq E[m]$ be a cyclic subgroup. Under the quotient map $\mathbb{C} \to E$, the subgroup C corresponds to a lattice $\Lambda_C \supseteq \Lambda$, namely

$$\Lambda_C = \{z \in \mathbb{C} \mid z + \Lambda \in C\}.$$

Define

$$\pi_C : \mathbb{C}/\Lambda \to \mathbb{C}/\Lambda_C, \quad z + \Lambda \mapsto z + \Lambda_C.$$

Then π_C is an isogeny with $\ker(\pi_C) = C$.

Proposition 11.2.22 *Let* $f : E = \mathbb{C}/\Lambda \to \mathbb{C}/\Lambda'$ *be an isogeny given by* $f(z+\Lambda) = mz + \Lambda'$, *and put* $K = \ker(f)$. *Then* f *is a composite of multiplication by a positive integer, a quotient by a cyclic subgroup and an isomorphism.*

Proof.

Step 1. Put $N = |K|$. Then $K \subseteq E[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$, so there exist $n, n' \in \mathbb{N}_{>0}$ such that

$$K \cong \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/(nn')\mathbb{Z}.$$

Multiplication by n maps K onto the cyclic subgroup $nK \cong \mathbb{Z}/n'\mathbb{Z}$.

Step 2. Write $\widetilde{K} = m^{-1}\Lambda'$ for the lattice corresponding to K, so $K = \widetilde{K}/\Lambda$. The lattice corresponding to nK is $n\widetilde{K}$. Consider the quotient isogeny

$$\pi : \mathbb{C}/\Lambda \to \mathbb{C}/(n\widetilde{K}), \quad z + \Lambda \mapsto z + n\widetilde{K},$$

whose kernel is nK.

Step 3. The map

$$\iota : \mathbb{C}/(n\widetilde{K}) \to \mathbb{C}/\Lambda', \quad z + n\widetilde{K} \mapsto \frac{m}{n}z + \Lambda'$$

is an isomorphism, since

$$\frac{m}{n}(n\widetilde{K}) = m\widetilde{K} = \Lambda'.$$

By construction,

$$\mathbb{C}/\Lambda \xrightarrow{[n]} \mathbb{C}/\Lambda \xrightarrow{\pi} \mathbb{C}/(n\widetilde{K}) \xrightarrow[\sim]{\iota} \mathbb{C}/\Lambda'$$

sends $z + \Lambda$ to $nz + \Lambda$, then to $nz + n\widetilde{K}$, and finally to $mz + \Lambda'$. Thus $f = \iota \circ \pi \circ [n]$.

□

Theorem-Definition 11.2.23 (Dual Isogeny) *Let $f : \mathbb{C}/\Lambda \to \mathbb{C}/\Lambda'$ be an isogeny given by $f(z + \Lambda) = mz + \Lambda'$. Choose a basis $\{\omega_1', \omega_2'\}$ of Λ' and positive integers m_1, m_2 such that $\{m_1\omega_1', m_2\omega_2'\}$ is a basis of $m\Lambda$. The dual isogeny of f is*

$$\widehat{f} : \mathbb{C}/\Lambda' \to \mathbb{C}/\Lambda, \quad z + \Lambda' \mapsto \frac{m_1 m_2}{m}z + \Lambda.$$

Moreover,

$$\deg(f) = m_1 m_2, \qquad \widehat{f} \circ f = [\deg(f)].$$

⌟

Proof. Since $m\Lambda \subseteq \Lambda'$, such bases and integers may be chosen. We have

$$m_1 m_2 \Lambda' \subseteq m\Lambda, \qquad \frac{m_1 m_2}{m}\Lambda' \subseteq \Lambda,$$

so the displayed formula defines an isogeny. Its composite with f satisfies

$$(\widehat{f} \circ f)(z + \Lambda) = \frac{m_1 m_2}{m} mz + \Lambda = m_1 m_2 z + \Lambda.$$

The map f has nonzero derivative in local coordinates, so each ramification index is 1. Its fibres are translates of its kernel, and

$$|\ker(f)| = [\Lambda' : m\Lambda] = m_1 m_2.$$

Hence $\deg(f) = m_1 m_2$ and $\widehat{f} \circ f = [\deg(f)]$. □

Note: Further properties of dual isogenies are left as exercises.

Definition 11.2.24 (Weil Pairing) Let $E = \mathbb{C}/\Lambda$ and $N \in \mathbb{N}_{>0}$. Choose a basis $\{\omega_1, \omega_2\}$ of Λ with $\omega_1/\omega_2 \in \mathcal{H}$. Then

$$E[N] \cong (\mathbb{Z}/N\mathbb{Z})^2 = \left\langle \frac{\omega_1}{N} + \Lambda \right\rangle \times \left\langle \frac{\omega_2}{N} + \Lambda \right\rangle.$$

For $P, Q \in E[N]$, write

$$P = a\frac{\omega_1}{N} + b\frac{\omega_2}{N} + \Lambda, \qquad Q = c\frac{\omega_1}{N} + d\frac{\omega_2}{N} + \Lambda,$$

with $a, b, c, d \in \mathbb{Z}/N\mathbb{Z}$, and put $\gamma_{P,Q} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Let

$$\mu_N = \{z \in \mathbb{C} \mid z^N = 1\} = \left\langle e^{2\pi i/N} \right\rangle.$$

The *Weil pairing* is

$$e_N : E[N] \times E[N] \to \mu_N, \quad (P, Q) \mapsto \exp\left(\frac{2\pi i}{N} \det(\gamma_{P,Q})\right),$$

where the determinant is evaluated using any integer representative modulo N. ⌟

Note: Verify that the Weil pairing is bilinear, alternating and nondegenerate, and check its compatibility as the torsion level varies, in particular for the maps $E[NM] \to E[N]$. These properties are left as exercises.

2.3. Elliptic Curves.

Let Λ be a lattice.

Definition 11.2.25 An *elliptic function* with respect to Λ is a meromorphic Λ-periodic function $f : \mathbb{C} \to \widehat{\mathbb{C}}$. By periodicity, it induces a map $\mathbb{C}/\Lambda \to \widehat{\mathbb{C}}$. ⌟

Lemma 11.2.26

(i) A holomorphic elliptic function is constant.

(ii) An elliptic function with no zeros is constant.

Proof.

(i) Let $f : \mathbb{C} \to \mathbb{C}$ be holomorphic and Λ-periodic. Choose a basis $\{\omega_1, \omega_2\}$ of Λ and fix $z_0 \in \mathbb{C}$. Consider the closed fundamental parallelogram

$$\overline{D} = \{z_0 + t_1\omega_1 + t_2\omega_2 \mid t_1, t_2 \in [0, 1]\}.$$

By periodicity,

$$\sup_{z \in \mathbb{C}} |f(z)| = \sup_{z \in \overline{D}} |f(z)|.$$

Since f is continuous and $\overline{D}$ is compact, the right-hand side is finite. Thus f is bounded and entire, so it is constant by Liouville's theorem.

(ii) If f has no zeros, then $1/f$ is holomorphic and Λ-periodic, with removable values at the poles of f. By (i), $1/f$ is constant, and hence f is constant.

□

Theorem 11.2.27 *Let f be an elliptic function with respect to Λ. Choose $z_0 \in \mathbb{C}$ and a basis $\{\omega_1, \omega_2\}$ of Λ, and put*

$$D = \{z_0 + t_1\omega_1 + t_2\omega_2 \mid t_1, t_2 \in [0, 1)\}.$$

Then:

(i) $\sum_{z\in D} \operatorname{res}_z(f) = 0.$

(ii) $\sum_{z\in D} \operatorname{ord}_z(f) = 0.$

⌟

Proof. We may choose the fundamental parallelogram so that f has no zeros or poles on ∂D.

(i) By the residue theorem,

$$\sum_{z\in D} \operatorname{res}_z(f) = \frac{1}{2\pi i}\int_{\partial D} f(z)\,\mathrm{d}z = 0.$$

Indeed, the integrals along opposite sides cancel by Λ-periodicity. For $f \equiv 0$, the conclusion is immediate.

(ii) Since f is Λ-periodic, so is f', and f'/f is an elliptic function. Applying (i) gives

$$\sum_{z\in D} \operatorname{res}_z\left(\frac{f'}{f}\right) = 0.$$

We have $\operatorname{res}_{z_0}(f'/f) = \operatorname{ord}_{z_0}(f)$. To see this, write locally

$$f(z) = (z - z_0)^m g(z),$$

where g is holomorphic, $g(z_0) \neq 0$ and $m = \operatorname{ord}_{z_0}(f)$. Then

$$\frac{f'(z)}{f(z)} = \frac{g'(z)}{g(z)} + \frac{m}{z - z_0}.$$

The first term is holomorphic at z_0, while the second has residue m. This proves (ii).

□

Definition 11.2.28 The *order* of an elliptic function f is the number of its poles in a fundamental domain D, counted with multiplicity. ⌟

Remark 11.2.29 By Theorem 11.2.27(ii), the order of a nonzero elliptic function also equals the number of its zeros in D, counted with multiplicity. ⌟

Corollary 11.2.30 *A nonconstant elliptic function has order at least* 2.

Proof.

Case 1 If the order is 0, then f is holomorphic, hence constant by Lemma 11.2.26(i).

Case 2 If the order is 1, then f has a single simple pole in D, say at z_0. By Theorem 11.2.27(i), $\operatorname{res}_{z_0}(f) = 0$, so this singularity is removable. Hence f is holomorphic and therefore constant.

Both cases contradict the assumption that f is nonconstant. □

Definition 11.2.31 The *Weierstrass $\wp$-function* with respect to Λ is

$$\wp_\Lambda(z) = \frac{1}{z^2} + \sum_{\omega\in\Lambda}{}' \left(\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2}\right).$$

Here the prime indicates that $\omega = 0$ is omitted. ⌟

Definition 11.2.32 Let $k > 2$ be an even integer. The *Eisenstein series of weight k with respect to Λ* is

$$G_k(\Lambda) = \sum_{\omega \in \Lambda}{}' \frac{1}{\omega^k}.$$

⌟

Note: By Exercise Sheet 2, the series $G_k(\Lambda)$ is absolutely convergent.

Remark 11.2.33 For $z \in \mathcal{H}$, consider the lattice $\Lambda_z = \mathbb{Z}z \oplus \mathbb{Z}1$. Then

$$G_k(\Lambda_z) = G_k(z),$$

where the right-hand side is the Eisenstein series introduced in §1.6.. ⌟

Proposition 11.2.34 *Let Λ be a lattice.*

(1) The series defining $\wp_\Lambda$ gives a meromorphic function on $\mathbb{C}$ with a double pole at every lattice point and no other poles.

(2) $\wp_\Lambda$ is an even elliptic function of order 2.

(3) Put $r = \min\{|\omega| \mid \omega \in \Lambda \setminus \{0\}\}$. For $0 < |z| < r$, its Laurent expansion is

$$\wp_\Lambda(z) = \frac{1}{z^2} + \sum_{n=1}^{\infty}(2n+1)G_{2n+2}(\Lambda)z^{2n}.$$

(4) The functions $\wp_\Lambda$ and $\wp'_\Lambda$ satisfy

$$\wp'_\Lambda(z)^2 = 4\wp_\Lambda(z)^3 - g_2(\Lambda)\wp_\Lambda(z) - g_3(\Lambda), \tag{11.2.1}$$

where $g_2(\Lambda) = 60G_4(\Lambda)$ and $g_3(\Lambda) = 140G_6(\Lambda)$. If $\{\omega_1, \omega_2\}$ is a basis of Λ, put $\omega_3 = \omega_1 + \omega_2$ and $e_i = \wp_\Lambda(\omega_i/2)$ for $i = 1, 2, 3$. Then the cubic on the right-hand side factors as

$$4x^3 - g_2(\Lambda)x - g_3(\Lambda) = 4(x - e_1)(x - e_2)(x - e_3).$$

(5) The cubic is nonsingular: $e_i \neq e_j$ whenever $i \neq j$. Its discriminant is

$$16\Delta(\Lambda) = 16\big(g_2(\Lambda)^3 - 27g_3(\Lambda)^2\big) \neq 0.$$

(6) Let E be the projective curve over $\mathbb{C}$ given in affine coordinates by

$$E : y^2 = 4x^3 - g_2(\Lambda)x - g_3(\Lambda).$$

The map

$$\varphi : \mathbb{C}/\Lambda \to E(\mathbb{C}) \subseteq \mathbb{P}^2(\mathbb{C}), \quad z + \Lambda \mapsto [\wp_\Lambda(z) : \wp'_\Lambda(z) : 1],$$

with $\varphi(0 + \Lambda) = [0 : 1 : 0]$, is biholomorphic.

(7) Given a curve $E : y^2 = 4x^3 - a_2x - a_3$ with $a_2^3 - 27a_3^2 \neq 0$, there exists a lattice Λ such that $a_2 = g_2(\Lambda)$ and $a_3 = g_3(\Lambda)$.

Proof.

(1) Note that

$$\frac{1}{(z-\omega)^2}-\frac{1}{\omega^2}=\frac{z(2\omega-z)}{\omega^2(z-\omega)^2}.$$

Fix $R>0$ and consider $\overline{D}_R=\{z\in\mathbb{C}\mid |z|\leqslant R\}$. Since $\overline{D}_R\cap\Lambda$ is finite, we split

$$\wp_\Lambda(z)=\frac{1}{z^2}+\sum_{\substack{\omega\in\Lambda\\ |\omega|\leqslant R}}{}'\left(\frac{1}{(z-\omega)^2}-\frac{1}{\omega^2}\right)+\sum_{\substack{\omega\in\Lambda\\ |\omega|>R}}\left(\frac{1}{(z-\omega)^2}-\frac{1}{\omega^2}\right).$$

It remains to estimate the tail uniformly for $|z|\leqslant R$. Write

$$R+d=\min\{|\omega|\mid \omega\in\Lambda,\ |\omega|>R\},\quad d>0.$$

Then

$$\left|\frac{z-\omega}{\omega}\right|=\left|1-\frac{z}{\omega}\right|\geqslant 1-\left|\frac{z}{\omega}\right|\geqslant 1-\frac{R}{R+d}.$$

Take

$$C=\left(1-\frac{R}{R+d}\right)^2.$$

The estimate on the board is

$$\left|\frac{1}{(z-\omega)^2}\right|\leqslant\frac{C}{|\omega|^2}.$$

Hence, for $|z|\leqslant R$ and $|\omega|>R$,

$$\left|\frac{z(2\omega-z)}{\omega^2(z-\omega)^2}\right|\leqslant\frac{CR(2|\omega|+R)}{|\omega|^4}\leqslant\frac{CR(2+R/|\omega|)}{|\omega|^3}\leqslant\frac{3CR}{|\omega|^3}.$$

The lattice sum $\sum_{\omega\neq0}|\omega|^{-3}$ converges, so the tail converges absolutely and uniformly on $\overline{D}_R$ and is holomorphic there. The finite part has a double pole at every point of $\Lambda\cap\overline{D}_R$. This proves (1).

(2) Since $\omega\mapsto-\omega$ permutes $\Lambda\setminus\{0\}$ and $(-z-\omega)^2=(z-(-\omega))^2$, the defining series gives $\wp_\Lambda(-z)=\wp_\Lambda(z)$.

To prove periodicity, differentiate the series:

$$\wp'_\Lambda(z)=-2\sum_{\omega\in\Lambda}\frac{1}{(z-\omega)^3}.$$

This is Λ-periodic. Therefore, for $\omega\in\Lambda$,

$$\wp_\Lambda(z+\omega)=\wp_\Lambda(z)+c(\omega).\tag{11.2.2}$$

Evaluate at $z=-\omega/2$. By evenness,

$$\wp_\Lambda(-\omega/2)=\wp_\Lambda(\omega/2)=\wp_\Lambda(-\omega/2)+c(\omega),$$

so $c(\omega)=0$. Thus $\wp_\Lambda$ is Λ-periodic. It has one double pole modulo Λ, so its order is 2.

(3) For $0 < |z| < r$ and $\omega \in \Lambda \setminus \{0\}$, we have $|z/\omega| < 1$. The geometric series gives

$$\frac{1}{(z-\omega)^2} = \frac{1}{\omega^2(1-z/\omega)^2} = \frac{1}{\omega^2}\left(1 + \sum_{m=1}^{\infty}(m+1)\left(\frac{z}{\omega}\right)^m\right).$$

Subtracting $1/\omega^2$ and summing over $\omega \neq 0$ yields

$$\wp_\Lambda(z) = \frac{1}{z^2} + \sum_{m=1}^{\infty}(m+1)\left(\sum_{\omega\neq 0}\frac{1}{\omega^{m+2}}\right)z^m.$$

Terms with odd m vanish by pairing ω and $-\omega$, giving the stated Laurent expansion.

(4) Near $z = 0$, the Laurent expansion gives

$$\wp_\Lambda(z) = \frac{1}{z^2} + 3G_4(\Lambda)z^2 + 5G_6(\Lambda)z^4 + \cdots,$$

and therefore

$$4\wp_\Lambda(z)^3 = \frac{4}{z^6} + \frac{36G_4(\Lambda)}{z^2} + 60G_6(\Lambda) + z(\cdots).$$

Also,

$$\wp'_\Lambda(z) = -\frac{2}{z^3} + 6G_4(\Lambda)z + 20G_6(\Lambda)z^3 + \cdots,$$

and the board records

$$\wp'_\Lambda(z)^2 = \frac{4}{z^6} + \frac{24G_4(\Lambda)}{z^2} - 80G_6(\Lambda) + z(\cdots).$$

The subsequent conclusion on the board is that

$$F(z) = \wp'_\Lambda(z)^2 - 4\wp_\Lambda(z)^3 + 60G_4(\Lambda)\wp_\Lambda(z) + 140G_6(\Lambda)$$

has a removable singularity at 0 and $F(0) = 0$. By periodicity, it is holomorphic everywhere. Hence F is a constant elliptic function, and this constant is 0. This gives (11.2.1).

Since $\wp_\Lambda$ is even, $\wp'_\Lambda$ is odd. For $i = 1, 2, 3$, periodicity gives

$$\wp'_\Lambda(\omega_i/2) = \wp'_\Lambda(-\omega_i/2) = -\wp'_\Lambda(\omega_i/2),$$

so $\wp'_\Lambda(\omega_i/2) = 0$. Substitution into (11.2.1) shows that the e_i are roots of the cubic.

(5) For each i, the elliptic function $f_i(z) = \wp_\Lambda(z) - e_i$ has order 2 and vanishes at $\omega_i/2$. Since $\wp'_\Lambda(\omega_i/2) = 0$, this zero has multiplicity at least 2, and thus is its only zero modulo Λ. The three nonzero half-periods $\omega_i/2$ are distinct modulo Λ, so for $j \neq i$,

$$0 \neq f_i(\omega_j/2) = e_j - e_i.$$

Hence the roots are distinct and the discriminant is nonzero.

(6) The board proves bijectivity as follows.

Step 1. Surjectivity. Let (x, y) be an affine point of $E(\mathbb{C})$. The elliptic function $\wp_\Lambda(z) - x$ is nonconstant, so it has a zero, say α. Since $\wp_\Lambda$ is even, $-\alpha$ is also a zero. By (11.2.1),

$$\wp'_\Lambda(\alpha)^2 = y^2.$$

Replacing α by $-\alpha$ if necessary, oddness of $\wp'_\Lambda$ gives $\wp'_\Lambda(\alpha) = y$. The point at infinity is the image of $0 + \Lambda$.

Step 2. Injectivity. Suppose $\varphi(z_1+\Lambda)=\varphi(z_2+\Lambda)$ for two nonzero classes. In particular,

$$\wp_\Lambda(z_1)=\wp_\Lambda(z_2),\qquad \wp'_\Lambda(z_1)=\wp'_\Lambda(z_2).$$

Case 1 $2z_1\notin\Lambda$. The order-2 elliptic function $\wp_\Lambda(z)-\wp_\Lambda(z_1)$ vanishes at z_1, $-z_1$ and z_2. The first two are distinct modulo Λ, so $z_2\equiv\pm z_1 \pmod\Lambda$. These zeros are simple, hence $\wp'_\Lambda(z_1)\neq 0$. Since $\wp'_\Lambda$ is odd and its values at z_1 and z_2 agree, the negative alternative is excluded. Thus $z_2\equiv z_1 \pmod\Lambda$.

Case 2 $2z_1\in\Lambda$. As above, $\wp_\Lambda(z)-\wp_\Lambda(z_1)$ has a double zero at z_1 and vanishes at z_2. Since its order is 2, we have $z_2\equiv z_1 \pmod\Lambda$.

Only the zero class maps to the point at infinity, so φ is injective.

(7) Let $E: y^2=4x^3-a_2x-a_3$ with $a_2^3-27a_3^2\neq 0$. We use the fact that $j:\mathcal{H}\to\mathbb{C}$ is surjective; this will be proved later. The cases $a_2=0$ or $a_3=0$ are left as exercises.

Suppose $a_2a_3\neq 0$. Choose $z\in\mathcal{H}$ such that

$$j(z)=1728\frac{a_2^3}{a_2^3-27a_3^2}.$$

Comparing with the definition of j gives

$$\frac{a_2^3}{g_2(z)^3}=\frac{a_3^2}{g_3(z)^2}. \tag{11.2.3}$$

Let $\omega_2\in\mathbb{C}^\times$, put $\omega_1=z\omega_2$ and $\Lambda=\mathbb{Z}\omega_1\oplus\mathbb{Z}\omega_2$. From the definition of the lattice Eisenstein series,

$$G_k(m\Lambda)=m^{-k}G_k(\Lambda),\qquad m\in\mathbb{C}^\times.$$

Therefore

$$g_2(\Lambda)=\omega_2^{-4}g_2(z),\qquad g_3(\Lambda)=\omega_2^{-6}g_3(z).$$

Choose ω_2 so that

$$\omega_2^{-4}=\frac{a_2}{g_2(z)},\qquad \omega_2^{-6}=\frac{a_3}{g_3(z)}.$$

Such a choice is possible by (11.2.3): after choosing a fourth root to satisfy the first equation, replacing ω_2 by $i\omega_2$ if necessary fixes the sign in the second. Thus $g_2(\Lambda)=a_2$ and $g_3(\Lambda)=a_3$.

□

Remark 11.2.35 The above construction gives the correspondence between complex tori and elliptic curves over $\mathbb{C}$. ⌟

Corollary 11.2.36 *For every $z\in\mathcal{H}$, $\Delta(z)\neq 0$.*

Proof. For $\Lambda_z=\mathbb{Z}z\oplus\mathbb{Z}1$, we have $\Delta(z)=\Delta(\Lambda_z)\neq 0$ by Proposition 11.2.34(5). □

2.4. Modular Curves and Moduli Spaces. Recall the correspondence between complex tori and elliptic curves over $\mathbb{C}$. Two elliptic curves $E = \mathbb{C}/\Lambda$ and $E' = \mathbb{C}/\Lambda'$ are isomorphic if and only if there exists $m \in \mathbb{C}^\times$ such that $m\Lambda = \Lambda'$. If $\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$ with $z = \omega_1/\omega_2 \in \mathcal{H}$, then

$$\Lambda_z = \mathbb{Z}z \oplus \mathbb{Z}1 = \frac{1}{\omega_2}\Lambda, \qquad E \cong E_z := \mathbb{C}/\Lambda_z.$$

Changing the oriented basis of Λ changes z by the action of $\mathrm{SL}_2(\mathbb{Z})$: if $\begin{pmatrix}\omega_1' \\ \omega_2'\end{pmatrix} = \gamma\begin{pmatrix}\omega_1 \\ \omega_2\end{pmatrix}$, then $\omega_1'/\omega_2' = \gamma z$.

Definition 11.2.37 Let $N \in \mathbb{N}_{>0}$.

1. Define

$$S_0(N) := \{(E, C) \mid E \text{ is an elliptic curve over } \mathbb{C},\ C \subseteq E \text{ is cyclic of order } N\}/\sim,$$

where $(E, C) \sim (E', C')$ if there exists an isomorphism $E \to E'$ carrying C onto C'. Write $[E, C]$ for the equivalence class of (E, C).

2. Define

$$S_1(N) := \{(E, Q) \mid E \text{ is an elliptic curve over } \mathbb{C},\ Q \in E \text{ has order } N\}/\sim,$$

where $(E, Q) \sim (E', Q')$ if there exists an isomorphism $E \to E'$ sending Q to Q'. Write $[E, Q]$ for the equivalence class of (E, Q).

3. Define

$$S(N) := \{(E, P, Q) \mid E \text{ is an elliptic curve over } \mathbb{C},\ E[N] = \langle P, Q\rangle,\ e_N(P, Q) = e^{2\pi i/N}\}/\sim .$$

Here the equivalence relation is given by isomorphisms of the marked elliptic curves. Write $[E, P, Q]$ for the equivalence class of (E, P, Q), where e_N is the Weil pairing from Definition 11.2.24.

⌟

Remark 11.2.38 For $N = 1$, the sets $S(1) = S_1(1) = S_0(1)$ are identified with the set of elliptic curves over $\mathbb{C}$ up to isomorphism. Write $[E]$ for the class of E. There are forgetful maps

$$S(N) \longrightarrow S_1(N) \longrightarrow S_0(N) \longrightarrow S(1),$$
$$[E, P, Q] \longmapsto [E, Q] \longmapsto [E, \langle Q\rangle] \longmapsto [E].$$

⌟

Definition 11.2.39 (Modular Curve) Let Γ be a congruence subgroup. The *modular curve* associated with Γ is

$$Y(\Gamma) := \Gamma\backslash\mathcal{H} = \{\Gamma z \mid z \in \mathcal{H}\}.$$

For $N \in \mathbb{N}_{>0}$, write

$$Y(N) := Y(\Gamma(N)), \qquad Y_1(N) := Y(\Gamma_1(N)), \qquad Y_0(N) := Y(\Gamma_0(N)).$$

⌟

Remark 11.2.40 For $N = 1$, we have $Y(1) = Y_1(1) = Y_0(1) = Y(\mathrm{SL}_2(\mathbb{Z}))$. The inclusions $\Gamma(N) \subseteq \Gamma_1(N) \subseteq \Gamma_0(N) \subseteq \mathrm{SL}_2(\mathbb{Z})$ induce maps

$$Y(N) \longrightarrow Y_1(N) \longrightarrow Y_0(N) \longrightarrow Y(1).$$

⌟

Theorem 11.2.41 *Let $N \in \mathbb{N}_{>0}$.*

1. *For every $[E, C] \in S_0(N)$, there exists $z \in \mathcal{H}$ such that*

$$[E, C] = \left[E_z, \left\langle \frac{1}{N} + \Lambda_z \right\rangle\right].$$

The map

$$\psi_0 : S_0(N) \longrightarrow Y_0(N), \quad \left[E_z, \left\langle \frac{1}{N} + \Lambda_z \right\rangle\right] \longmapsto \Gamma_0(N)z$$

is a bijection.

2. *For every $[E, Q] \in S_1(N)$, there exists $z \in \mathcal{H}$ such that*

$$[E, Q] = \left[E_z, \frac{1}{N} + \Lambda_z\right].$$

The map

$$\psi_1 : S_1(N) \longrightarrow Y_1(N), \quad \left[E_z, \frac{1}{N} + \Lambda_z\right] \longmapsto \Gamma_1(N)z$$

is a bijection.

3. *For every $[E, P, Q] \in S(N)$, there exists $z \in \mathcal{H}$ such that*

$$[E, P, Q] = \left[E_z, \frac{z}{N} + \Lambda_z, \frac{1}{N} + \Lambda_z\right].$$

The map

$$\psi : S(N) \longrightarrow Y(N), \quad \left[E_z, \frac{z}{N} + \Lambda_z, \frac{1}{N} + \Lambda_z\right] \longmapsto \Gamma(N)z$$

is a bijection.

⌟

Proof. We first prove (2).

Step 1. Let $[E, Q] \in S_1(N)$. Choose $z \in \mathcal{H}$ and an isomorphism $E \cong E_z$. The image of Q has the form

$$\frac{cz + d}{N} + \Lambda_z, \qquad c, d \in \mathbb{Z}.$$

Since Q has order N, we have $\gcd(c, d, N) = 1$. By Bézout's identity, there exist $a, b, k \in \mathbb{Z}$ such that $ad - bc - kN = 1$. Thus $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ reduces to an element of $\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$. Since the reduction map $\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$ is surjective, we may replace γ by a lift in $\mathrm{SL}_2(\mathbb{Z})$ without changing $(cz + d)/N + \Lambda_z$.

Put $z' = \gamma z$ and $m = cz + d$. Then $mz' = az + b$, and

$$m\Lambda_{z'} = \mathbb{Z}(az+b) \oplus \mathbb{Z}(cz+d) = \Lambda_z, \qquad m\left(\frac{1}{N} + \Lambda_{z'}\right) = \frac{cz+d}{N} + \Lambda_z.$$

Hence $[E, Q] = [E_{z'}, 1/N + \Lambda_{z'}]$.

Step 2. We check that the maps

$$S_1(N) \longrightarrow Y_1(N), \quad [E_z, 1/N + \Lambda_z] \longmapsto \Gamma_1(N)z,$$

and

$$Y_1(N) \longrightarrow S_1(N), \quad \Gamma_1(N)z \longmapsto [E_z, 1/N + \Lambda_z]$$

are well-defined and injective.

Let $\Gamma_1(N)z = \Gamma_1(N)z'$. Choose $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_1(N)$ with $\gamma z' = z$. The board gives $m = cz + d$ and

$$m\Lambda_z = \Lambda_{z'}, \qquad m\left(\frac{1}{N} + \Lambda_z\right) = \frac{cz'+d}{N} + \Lambda_{z'}.$$

Since $(c, d) \equiv (0, 1) \pmod N$, the latter point equals $1/N + \Lambda_{z'}$. Thus $[E_z, 1/N + \Lambda_z] = [E_{z'}, 1/N + \Lambda_{z'}]$.

Conversely, suppose $[E_z, 1/N + \Lambda_z] = [E_{z'}, 1/N + \Lambda_{z'}]$. Then there exists $m \in \mathbb{C}^\times$ such that

$$m\Lambda_z = \Lambda_{z'}, \qquad m\left(\frac{1}{N} + \Lambda_z\right) = \frac{1}{N} + \Lambda_{z'}.$$

Comparing oriented bases gives $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ such that

$$\begin{pmatrix} mz \\ m \end{pmatrix} = g \begin{pmatrix} z' \\ 1 \end{pmatrix}.$$

In particular, $m = cz' + d$ and $z = gz'$. The condition on the marked point gives

$$\frac{1}{N} + \Lambda_{z'} = \frac{cz'+d}{N} + \Lambda_{z'} \Longrightarrow c \equiv 0 \pmod N, \quad d \equiv 1 \pmod N.$$

Thus $g \in \Gamma_1(N)$ and $\Gamma_1(N)z = \Gamma_1(N)z'$. The two maps are inverse bijections.

For (1), choose a generator Q of C and apply (2) to obtain $[E, C] = [E_z, \langle 1/N + \Lambda_z \rangle]$. The verification of the bijection follows the same argument, replacing equality of marked points by equality of the cyclic subgroups they generate. If $\gamma \in \Gamma_0(N)$, then $c \equiv 0 \pmod N$ and $d \bmod N \in (\mathbb{Z}/N\mathbb{Z})^\times$. Consequently,

$$\frac{cz'+d}{N} + \Lambda_{z'} = \frac{d}{N} + \Lambda_{z'}, \qquad \left\langle \frac{d}{N} + \Lambda_{z'} \right\rangle = \left\langle \frac{1}{N} + \Lambda_{z'} \right\rangle.$$

Conversely, equality of the cyclic subgroups implies $c \equiv 0 \pmod N$ and $d \bmod N \in (\mathbb{Z}/N\mathbb{Z})^\times$. The final line on the board reads $\gamma \in \Gamma_1(N)$.

The proof of (3) is not included in these lecture notes. □

2.5. Modular Curves as Topological Spaces. Let Γ be a congruence subgroup. We equip $Y(\Gamma) = \Gamma\backslash\mathcal{H}$ with the quotient topology for

$$\pi : \mathcal{H} \longrightarrow Y(\Gamma), \quad z \longmapsto \Gamma z.$$

Thus $U \subseteq Y(\Gamma)$ is open if and only if $\pi^{-1}(U)$ is open in $\mathcal{H}$.

Remark 11.2.42 For $U_1, U_2 \subseteq \mathcal{H}$,

$$\begin{aligned}\pi(U_1) \cap \pi(U_2) = \varnothing &\iff \pi^{-1}(\pi(U_1)) \cap \pi^{-1}(\pi(U_2)) = \varnothing \\ &\iff \Gamma U_1 \cap \Gamma U_2 = \varnothing \\ &\iff \Gamma U_1 \cap U_2 = \varnothing.\end{aligned}$$

⌟

Proposition 11.2.43 *The quotient map $\pi : \mathcal{H} \to Y(\Gamma)$ is open.*

Proof. Let $V \subseteq \mathcal{H}$ be open. Then

$$\pi^{-1}(\pi(V)) = \Gamma V = \{\gamma v \mid \gamma \in \Gamma,\ v \in V\} = \bigcup_{\gamma\in\Gamma} \gamma V.$$

Each γ acts by a homeomorphism on $\mathcal{H}$, so each γV is open. Hence $\pi^{-1}(\pi(V))$ is open, and therefore $\pi(V)$ is open in $Y(\Gamma)$. □

Proposition 11.2.44 *For every $z_1, z_2 \in \mathcal{H}$, there exist neighbourhoods $U_1 \ni z_1$ and $U_2 \ni z_2$ such that, for every $\gamma \in \mathrm{SL}_2(\mathbb{Z})$,*

$$\gamma U_1 \cap U_2 \neq \varnothing \Longrightarrow \gamma z_1 = z_2.$$

Lemma 11.2.45 *Fix $c, d \in \mathbb{Z}$ and $\begin{pmatrix}\alpha & \beta \\ c & d\end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$. Then*

$$\left\{\begin{pmatrix}a & b \\ c & d\end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})\right\} = \left\{\begin{pmatrix}1 & k \\ 0 & 1\end{pmatrix}\begin{pmatrix}\alpha & \beta \\ c & d\end{pmatrix} \,\middle|\, k \in \mathbb{Z}\right\}.$$

Proof. Let $\begin{pmatrix}a & b \\ c & d\end{pmatrix}, \begin{pmatrix}\alpha & \beta \\ c & d\end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$. Then $ad - bc = \alpha d - \beta c = 1$. The board records

$$bc \equiv 1 \equiv \beta c \pmod{d}, \qquad ad \equiv 1 \equiv \alpha d \pmod{c},$$

and obtains $b \equiv \beta \pmod{d}$ and $a \equiv \alpha \pmod{c}$. Write $\beta = b + md$ and $\alpha = a + nc$. Comparing determinants gives

$$1 = (a + nc)d - (b + md)c = ad - bc + (n - m)cd.$$

The board concludes $m = n$, and hence

$$\begin{pmatrix}\alpha & \beta \\ c & d\end{pmatrix} = \begin{pmatrix}a + mc & b + md \\ c & d\end{pmatrix} = \begin{pmatrix}1 & m \\ 0 & 1\end{pmatrix}\begin{pmatrix}a & b \\ c & d\end{pmatrix}.$$

□

Proof of Proposition 11.2.44.

Step 1. Choose open discs $U_1' \ni z_1$ and $U_2' \ni z_2$ whose closures are compact subsets of $\mathcal{H}$. Put

$$y_1 = \inf_{z \in U_1'} \mathrm{Im}(z) > 0, \qquad y_2 = \inf_{z \in U_2'} \mathrm{Im}(z) > 0, \qquad Y_1 = \sup_{z \in U_1'} \mathrm{Im}(z) < \infty.$$

For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ and $z \in U_1'$, recall that

$$\mathrm{Im}(\gamma z) = \frac{\mathrm{Im}(z)}{|cz+d|^2} = \frac{\mathrm{Im}(z)}{(c\Re(z)+d)^2 + (c\mathrm{Im}(z))^2}.$$

The board uses the estimate

$$\mathrm{Im}(\gamma z) \leq \min\left\{ \frac{Y_1}{(cy_1)^2}, \frac{Y_1}{(c\Re(z)+d)^2} \right\}.$$

The first bound is less than y_2 for all but finitely many $c \in \mathbb{Z}$. For each remaining c, the second bound is less than y_2 for all but finitely many $d \in \mathbb{Z}$, uniformly for $z \in U_1'$. Thus $\gamma U_1' \cap U_2'$ can be nonempty for only finitely many bottom rows (c, d).

Step 2. For each such bottom row, Lemma 11.2.45 gives a fixed matrix γ_0 such that every matrix with that bottom row is $T^k \gamma_0$, where $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $k \in \mathbb{Z}$. Hence

$$\gamma U_1' \cap U_2' = (\gamma_0 U_1' + k) \cap U_2'.$$

Since the discs have compact closures in $\mathcal{H}$, this intersection is empty for all but finitely many k. Therefore the set

$$F := \{\gamma \in \mathrm{SL}_2(\mathbb{Z}) \mid \gamma U_1' \cap U_2' \neq \varnothing,\ \gamma z_1 \neq z_2\}$$

is finite.

Step 3. Since $\mathcal{H}$ is Hausdorff, for every $\gamma \in F$ choose disjoint neighbourhoods $U_{1,\gamma} \ni \gamma z_1$ and $U_{2,\gamma} \ni z_2$. Put

$$U_1 = U_1' \cap \bigcap_{\gamma \in F} \gamma^{-1} U_{1,\gamma}, \qquad U_2 = U_2' \cap \bigcap_{\gamma \in F} U_{2,\gamma}.$$

These are neighbourhoods of z_1 and z_2. If $\gamma U_1 \cap U_2 \neq \varnothing$ and $\gamma z_1 \neq z_2$, then $\gamma \in F$. But then $\gamma U_1 \subseteq U_{1,\gamma}$ and $U_2 \subseteq U_{2,\gamma}$, contradicting $U_{1,\gamma} \cap U_{2,\gamma} = \varnothing$. Thus $\gamma z_1 = z_2$.

□

Proposition 11.2.46 *The space $Y(\Gamma)$ is Hausdorff.*

Proof. Let $x, y \in Y(\Gamma)$ with $x \neq y$. Choose $z_1, z_2 \in \mathcal{H}$ with $\pi(z_1) = x$ and $\pi(z_2) = y$, and choose U_1, U_2 as in Proposition 11.2.44. Since $x \neq y$, we have $\gamma z_1 \neq z_2$ for every $\gamma \in \Gamma$. Hence $\Gamma U_1 \cap U_2 = \varnothing$, so Remark 11.2.42 gives $\pi(U_1) \cap \pi(U_2) = \varnothing$. By Proposition 11.2.43, the sets $\pi(U_1)$ and $\pi(U_2)$ are neighbourhoods of x and y, respectively. □

2.6. Modular Curves as Riemann Surfaces. Let Γ be a congruence subgroup and $\pi : \mathcal{H} \to Y(\Gamma)$ the quotient map. Recall that $\mathrm{SL}_2(\mathbb{Z})/\{\pm\mathrm{Id}\}$ acts faithfully on $\mathcal{H}$. For $z \in \mathcal{H}$, put

$$\Gamma_z := \{\gamma \in \Gamma \mid \gamma z = z\},$$

the *stabilizer* of z.

Suppose first that $\Gamma_z = \Gamma \cap \{\pm\mathrm{Id}\}$. Apply Proposition 11.2.44 with $z_1 = z_2 = z$ and put $U = U_1 \cap U_2$. Then

$$\gamma U \cap U \neq \varnothing \Longrightarrow \gamma z = z \Longrightarrow \gamma \in \Gamma \cap \{\pm\mathrm{Id}\}.$$

The restriction $\pi|_U : U \to \pi(U)$ is a homeomorphism. Indeed, if $z_1, z_2 \in U$ satisfy $\pi(z_1) = \pi(z_2)$, there exists $\gamma \in \Gamma$ with $\gamma z_1 = z_2$. Thus $\gamma U \cap U \neq \varnothing$, so $\gamma \in \{\pm\mathrm{Id}\}$ and $z_1 = z_2$. Since π is continuous and open, its restriction is a homeomorphism.

Definition 11.2.47 (Elliptic Point) A point $z \in \mathcal{H}$ is *elliptic* with respect to Γ if $\Gamma_z \neq \Gamma \cap \{\pm\mathrm{Id}\}$. Its image $\pi(z) \in Y(\Gamma)$ is also called an elliptic point. ⌟

Example 11.2.48 For $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, the point i is elliptic, and

$$\mathrm{SL}_2(\mathbb{Z})_i = \left\langle S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right\rangle.$$

Proposition 11.2.49 *If $z \in \mathcal{H}$ is elliptic, then Γ_z is a finite cyclic group.*

Definition 11.2.50 (Period) The *period* of $z \in \mathcal{H}$ with respect to Γ is

$$h_z(\Gamma) := |\Gamma_z/(\Gamma_z \cap \{\pm\mathrm{Id}\})| = \begin{cases} |\Gamma_z|/2, & -\mathrm{Id} \in \Gamma, \\ |\Gamma_z|, & -\mathrm{Id} \notin \Gamma. \end{cases}$$

We often write $h_z = h_z(\Gamma)$ when Γ is clear from the context. ⌟

Remark 11.2.51 A point z is elliptic if and only if $h_z > 1$. For every $\gamma \in \mathrm{SL}_2(\mathbb{Z})$,

$$h_z(\Gamma) = h_{\gamma z}(\gamma \Gamma \gamma^{-1}).$$

Indeed, conjugation gives an isomorphism

$$\Gamma_z \longrightarrow (\gamma\Gamma\gamma^{-1})_{\gamma z}, \quad g \longmapsto \gamma g \gamma^{-1}.$$

Since $\{\pm\mathrm{Id}\}$ is the centre of $\mathrm{SL}_2(\mathbb{Z})$, we have $-\mathrm{Id} \in \Gamma$ if and only if $-\mathrm{Id} \in \gamma\Gamma\gamma^{-1}$. In particular, $h_z(\Gamma)$ depends only on the orbit Γz, so it is well-defined for $\pi(z) \in Y(\Gamma)$. ⌟

Remark 11.2.52 If $\gamma \in \Gamma_z$, then $\gamma\overline{z} = \overline{z}$, viewing γ as a fractional linear transformation on the extended complex plane. ⌟

Corollary 11.2.53 *For every $z \in \mathcal{H}$, there exists a neighbourhood $U \ni z$ such that*

$$\gamma U \cap U \neq \varnothing \Longrightarrow \gamma \in \Gamma_z \qquad (\gamma \in \Gamma),$$

and U has no elliptic points except possibly z.

Proof. Apply Proposition 11.2.44 with $z_1 = z_2 = z$ and put $U = U_1 \cap U_2$. The implication follows immediately. Suppose $z' \in U$, $z' \neq z$, and both z and z' are elliptic. For $\gamma \in \Gamma_{z'}$, the equality $\gamma z' = z'$ implies $\gamma U \cap U \neq \varnothing$, and hence $\gamma \in \Gamma_z$. Write $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Since $\gamma z = z$ and $\gamma z' = z'$, the polynomial

$$P(X) = cX^2 + (d-a)X - b \in \mathbb{Z}[X]$$

vanishes at $z, z', \overline{z}, \overline{z'}$. These are four distinct points, so $P = 0$. Hence $c = b = 0$ and $a = d$, giving $\gamma \in \{\pm \mathrm{Id}\}$. The board concludes that this contradicts ellipticity of z'. □

For $z \in \mathcal{H}$, put

$$\delta_z = \begin{pmatrix} 1 & -z \\ 1 & -\overline{z} \end{pmatrix} \in \mathrm{GL}_2(\mathbb{C}).$$

Then $\delta_z(z) = 0$ and $\delta_z(\overline{z}) = \infty$. Conjugation identifies the stabilizer of 0 in the transformed group with

$$\left(\delta_z\{\pm\mathrm{Id}\}\Gamma\delta_z^{-1}\right)_0 / \{\pm\mathrm{Id}\} \cong \left(\delta_z\{\pm\mathrm{Id}\}\Gamma_z\delta_z^{-1}\right) / \{\pm\mathrm{Id}\},$$

which is finite cyclic of order h_z. Every element fixes both 0 and ∞, and hence acts by multiplication:

$$w \longmapsto \alpha w, \qquad \alpha = e^{2\pi i k/h_z}, \quad k \in \mathbb{Z}.$$

Let $U \ni z$ be a neighbourhood as in Corollary 11.2.53. Define

$$\psi := \rho_{h_z} \circ \delta_z : U \longrightarrow \mathbb{C}, \qquad \rho_{h_z}(w) = w^{h_z}.$$

The image $\psi(U)$ is open by the open mapping theorem. For $u, v \in U$,

$$\begin{aligned} \pi(u) = \pi(v) &\iff \text{there exists } \gamma \in \Gamma \text{ with } \gamma u = v \\ &\iff \text{there exists } \gamma \in \Gamma_z \text{ with } \gamma u = v \\ &\iff \text{there exists } k \in \mathbb{Z} \text{ with } e^{2\pi i k/h_z}\delta_z(u) = \delta_z(v) \\ &\iff \delta_z(u)^{h_z} = \delta_z(v)^{h_z} \\ &\iff \psi(u) = \psi(v). \end{aligned}$$

Thus ψ induces a bijection $\varphi : \pi(U) \to \psi(U)$ with $\varphi \circ \pi|_U = \psi$:

$$\begin{array}{ccc} U & \xrightarrow{\ \pi\ } & \pi(U) \\ {\scriptstyle\psi}\downarrow & \swarrow_{\varphi} & \\ \psi(U) & & \end{array}$$

Since ψ and $\pi|_U$ are continuous and open, φ is a homeomorphism. More explicitly, for $V \subseteq \psi(U)$ open and $W \subseteq \pi(U)$ open,

$$\varphi^{-1}(V) = \pi(\psi^{-1}(V)), \qquad \varphi(W) = \psi\big((\pi|_U)^{-1}(W)\big)$$

are open. This gives a local chart at $\pi(z)$ for every $z \in \mathcal{H}$.

Remark 11.2.54 If $h_z = 1$, this chart agrees with the initial construction up to the change of coordinates δ_z. ⌟

Note: The compactification $X(\Gamma)$ will be obtained by adding the cusps.

Proposition 11.2.55 *The local charts constructed above are compatible and define a Riemann surface structure on $Y(\Gamma)$.*

Proof. Let $(\pi(U_1), \varphi_1)$ and $(\pi(U_2), \varphi_2)$ be two such charts, centred at the images of z_1^0 and z_2^0, respectively. Write $\delta_j = \delta_{z_j^0}$ and $h_j = h_{z_j^0}$. We must check that

$$\varphi_{21} := \varphi_2 \circ \varphi_1^{-1} : V_{12} \longrightarrow V_{21}, \qquad V_{12} = \varphi_1(\pi(U_1) \cap \pi(U_2)), \quad V_{21} = \varphi_2(\pi(U_1) \cap \pi(U_2)),$$

is biholomorphic. Fix $x = \pi(z_1) = \pi(z_2)$ in the overlap, with $z_j \in U_j$, and choose $\gamma \in \Gamma$ with $\gamma z_1 = z_2$. Put $U_{12} = U_1 \cap \gamma^{-1} U_2$. For $z \in U_{12}$ and $q = \varphi_1(\pi(z)) = \delta_1(z)^{h_1}$, we have

$$\varphi_{21}(q) = \varphi_2(\pi(\gamma z)) = \delta_2(\gamma z)^{h_2}.$$

Case 1. $\varphi_1(x) = 0$ and $h_1 > 1$. Then $z_1 = z_1^0$ is elliptic. Since $z_2 = \gamma z_1$ is also elliptic and U_2 contains no elliptic point other than possibly its centre, $z_2 = z_2^0$ and $h_2 = h_1$. Moreover, $\delta_2 \gamma \delta_1^{-1}$ fixes 0 and ∞, since it sends

$$0 \longmapsto z_1 \longmapsto z_2 \longmapsto 0, \qquad \infty \longmapsto \overline{z_1} \longmapsto \overline{z_2} \longmapsto \infty.$$

Thus it is represented by $\begin{pmatrix} \alpha & 0 \\ 0 & \beta \end{pmatrix}$ with $\alpha, \beta \in \mathbb{C}^\times$, and

$$\varphi_{21}(q) = \left(\frac{\alpha}{\beta}\right)^{h_1} q.$$

Hence the transition map is biholomorphic near $\varphi_1(x)$.

Case 2. $\varphi_1(x) \neq 0$, or $h_1 = 1$. The derivative of $z \mapsto \delta_1(z)^{h_1}$ at z_1 is nonzero. Indeed, $\delta_1'(z_1) \neq 0$, and either $h_1 = 1$ or $\delta_1(z_1) \neq 0$. It therefore has a holomorphic local inverse η near $q_0 = \varphi_1(x)$. On a sufficiently small neighbourhood of q_0,

$$\varphi_{21}(q) = \delta_2(\gamma \eta(q))^{h_2},$$

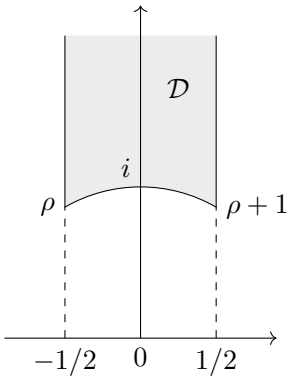
so φ_{21} is holomorphic. If $\varphi_2(x) = 0$ and $h_2 > 1$, then x is elliptic; the isolation of elliptic points in U_1 would force $z_1 = z_1^0$ and $h_1 > 1$, contrary to this case. Thus the same local inverse argument applies to φ_{12}.

The transition maps are biholomorphic throughout their domains. □

Put $Y = Y(\mathrm{SL}_2(\mathbb{Z}))$ and

$$\mathcal{D} := \{z \in \mathcal{H} \mid |\Re(z)| \leq 1/2, \ |z| \geq 1\}, \qquad \rho = e^{2\pi i/3}.$$

The two endpoints of the circular boundary are ρ and $\rho + 1 = -\overline{\rho} = e^{\pi i/3}$.



Proposition 11.2.56

1. *The restriction $\pi : \mathcal{D} \to Y$ is surjective.*
2. *If $z_1, z_2 \in \mathcal{D}$ are distinct and $\gamma z_1 = z_2$ for some $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, then either $\Re(z_1) = \pm 1/2$ and $z_2 = z_1 \mp 1$, or $|z_1| = 1$ and $z_2 = -1/z_1$.*
3. *The elliptic points in $\mathcal{H}$ for $\mathrm{SL}_2(\mathbb{Z})$ form the two orbits $\mathrm{SL}_2(\mathbb{Z})i$ and $\mathrm{SL}_2(\mathbb{Z})\rho$. Their stabilizers are*
$$\mathrm{SL}_2(\mathbb{Z})_i = \langle S \rangle, \qquad \mathrm{SL}_2(\mathbb{Z})_\rho = \langle ST \rangle,$$
of orders 4 and 6, respectively, where
$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad ST = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}.$$

Proof.

1. Fix $z \in \mathcal{H}$. For every $A > 0$, the set $\{(c,d) \in \mathbb{Z}^2 \mid |cz+d| < A\}$ is finite: $|c|\mathrm{Im}(z) < A$ bounds c, and then $|c\Re(z) + d| < A$ bounds d. Since
$$\mathrm{Im}(\gamma z) = \frac{\mathrm{Im}(z)}{|cz+d|^2},$$
there exists $\gamma' \in \mathrm{SL}_2(\mathbb{Z})$ for which $\mathrm{Im}(\gamma' z)$ is maximal. Choose $n \in \mathbb{Z}$ such that $w = T^n \gamma' z$ satisfies $|\Re(w)| \le 1/2$. Translation does not change the imaginary part, so w still has maximal imaginary part. If $|w| < 1$, then
$$\mathrm{Im}(Sw) = \frac{\mathrm{Im}(w)}{|w|^2} > \mathrm{Im}(w),$$
a contradiction. Hence $w \in \mathcal{D}$.

2. We prove (2) and (3) together. Let $z \in \mathcal{D}$ and $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ with $\gamma z \in \mathcal{D}$. Exchanging (z, γ) with $(\gamma z, \gamma^{-1})$ if necessary, assume $\mathrm{Im}(\gamma z) \ge \mathrm{Im}(z)$. Writing $z = x + iy$, this gives
$$|cz+d|^2 = (cx+d)^2 + c^2 y^2 \le 1.$$
Since $z \in \mathcal{D}$, we have $y^2 \ge 1 - x^2 \ge 3/4$, and hence $c \in \{-1, 0, 1\}$.

Case 1. $c = 0$. Then $a = d = \pm 1$, so γ acts by an integral translation. Since z and γz lie in the strip $|\Re(w)| \leq 1/2$, either $\gamma = \pm \mathrm{Id}$, or the translation is by ± 1 and exchanges the two vertical boundaries.

Case 2. $c = 1$. The inequality $|z + d| \leq 1$ forces $d = 0$, except that $d = 1$ is also possible for $z = \rho$ and $d = -1$ for $z = \rho + 1$. If $d = 0$, then $b = -1$ and $\gamma z = a - 1/z$. The conditions $|z| \geq 1$ and $|z| \leq 1$ give $|z| = 1$, so $-1/z = -\overline{z}$. Thus $a = 0$ except at the endpoints: for $z = \rho$, $a = 0, -1$, and for $z = \rho + 1$, $a = 0, 1$. These give either the circular boundary identification or a fixed endpoint.

If $z = \rho$ and $d = 1$, then $a - b = 1$ and

$$\gamma\rho = a - \frac{1}{\rho + 1} = a + \rho.$$

Hence $a = 0, 1$. Similarly, for $z = \rho + 1$ and $d = -1$, we obtain $\gamma z = a + z$ with $a = 0, -1$. These again give a fixed endpoint or the identification of the two endpoints.

Case 3. $c = -1$. Replace γ by $-\gamma$, which has the same action on $\mathcal{H}$, and apply the preceding case.

This proves (2). The same classification with $\gamma z = z$ shows that, inside $\mathcal{D}$, a noncentral stabilizer is possible only at i, ρ, or $\rho + 1$. At i, the matrices are $\{\pm \mathrm{Id}, \pm S\} = \langle S \rangle$. At ρ, they are $\{\pm \mathrm{Id}, \pm ST, \pm (ST)^2\} = \langle ST \rangle$. The point $\rho + 1$ is equivalent to ρ by T, so its stabilizer is conjugate to that of ρ. Part (1) now gives (3) for all of $\mathcal{H}$.

□

Remark 11.2.57 Parts (1) and (2) say that $\mathcal{D}$ is a fundamental domain for the action of $\mathrm{SL}_2(\mathbb{Z})$ on $\mathcal{H}$. ⌟

Corollary 11.2.58 *Let Γ be a congruence subgroup. Then $Y(\Gamma)$ has finitely many elliptic points, and every elliptic stabilizer Γ_z is finite cyclic.*

Proof. For every $z \in \mathcal{H}$, we have $\Gamma_z \subseteq \mathrm{SL}_2(\mathbb{Z})_z$. By Proposition 11.2.56, the latter is finite cyclic, and hence so is Γ_z. This also proves Proposition 11.2.49.

Write $\mathrm{SL}_2(\mathbb{Z}) = \bigcup_{j=1}^{r} \Gamma \delta_j$, where $r = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma] < \infty$. Every point elliptic for Γ is elliptic for $\mathrm{SL}_2(\mathbb{Z})$, so its Γ-orbit is among

$$\{\Gamma\delta_j i,\ \Gamma\delta_j\rho \mid 1 \leq j \leq r\}.$$

There are at most $2r$ such orbits. □

2.7. The Compact Modular Curves. Let $\mathcal{H}^* := \mathcal{H} \cup \mathbb{Q} \cup \{\infty\}$. Recall that $\mathrm{SL}_2(\mathbb{Z})$ acts transitively on $\mathbb{Q} \cup \{\infty\}$, and

$$\mathrm{SL}_2(\mathbb{Z})_\infty = \left\{ \pm \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \;\middle|\; n \in \mathbb{Z} \right\}.$$

Definition 11.2.59 (Compact Modular Curve) The *compact modular curve* associated with a congruence subgroup Γ is

$$X(\Gamma) := \Gamma\backslash\mathcal{H}^* = Y(\Gamma) \cup \big(\Gamma\backslash(\mathbb{Q} \cup \{\infty\})\big).$$

Thus $X(\Gamma)$ is obtained from $Y(\Gamma)$ by adding the finitely many cusps of Γ. For $N \in \mathbb{N}_{>0}$, write

$$X(N) := X(\Gamma(N)), \qquad X_1(N) := X(\Gamma_1(N)), \qquad X_0(N) := X(\Gamma_0(N)).$$

⌟

For $M \in \mathbb{R}_{>0}$, put

$$\mathcal{N}_M := \{z \in \mathcal{H} \mid \mathrm{Im}(z) > M\}.$$

Equip $\mathcal{H}^*$ with the topology generated by the open subsets of $\mathcal{H}$ and the sets

$$\alpha\big(\mathcal{N}_M \cup \{\infty\}\big), \qquad \alpha \in \mathrm{SL}_2(\mathbb{Z}),\ M > 0.$$

Every $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ acts on $\mathcal{H}^*$ by a homeomorphism. We equip $X(\Gamma)$ with the quotient topology for the natural projection $\pi^* : \mathcal{H}^* \to X(\Gamma)$.

Note: In the next lecture, we will prove that $X(\Gamma)$ is Hausdorff, connected and compact.